# Aspirations, Hope, and Savings for Old Age: Evidence from a Field Experiment in Ghana*

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**Abstract**

This study evaluates the effectiveness of a motivational video and a savings calendar in encouraging smallholder cocoa farmers in rural Ghana to save more for old age, contributing to the growing body of evidence on how psychological and behavioral interventions can promote long-term savings among the rural poor. The most significant result is that the combination of the video and the savings calendar led to a statistically significant increase in the fraction of farmers saving for old age in the long run. The intervention's impact on the amount saved was particularly pronounced among farmers who already had higher levels of savings at baseline. While the video successfully raised aspirations in the short term, its effects on aspirations faded over time. Moreover, while the intervention increased savings participation overall, its effects were not equally distributed. Women, in particular, did not benefit as much as men in terms of agency and hope, despite the savings calendar's intent to make goals more salient.

**Keywords**: Aspirations; Hope; Pension Savings; Behavioral Economics; Field Experiment; Ghana

**JEL Classification**: D91; O12; O16; G51

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# 1 Introduction

Poverty among older adults is an increasingly serious challenge in many developing countries (e.g., Banerjee et al., 2024), especially those in sub-Saharan Africa (SSA). While traditional social security systems, such as family- and community-based safety nets, are eroding (Duhon et al., 2025; Kos and Lensink, 2023), SSA has recorded the world's largest gains in life expectancy over the past decade and is projected to continue doing so until the 2040s; consequently, the region's older population is expected to rise from around 52 million to roughly 158 million by 2050 (United Nations Population Division, 2022).

In light of these developments, pension schemes and other long-term savings mechanisms aimed at enhancing the well-being of older adults are crucial (Huang and Zhang, 2021; Kolsrud et al., 2024; Laitner and Silverman, 2012). Yet, although several countries in the region have introduced public social protection programs, including social pensions, pension coverage remains alarmingly low—at only 19.8% in sharp contrast to the global average of 77.5% (International Labour Organization, 2021). Among the poor, uptake and use of long-term savings accounts and pension products offered by the private sector also remain very limited. For smallholders, who face large income fluctuations and are especially vulnerable to shocks, accumulating long-term savings is particularly challenging.

Research on micro-savings products–financial tools explicitly tailored to help low-income individuals, including smallholders–shows that removing external barriers to accessing savings accounts, such as high account-opening fees or complicated procedures, significantly increases uptake. This indicates a high latent demand for formal savings among the poor (e.g., Dupas and Robinson, 2013; Karlan et al., 2014; Pomeranz and Kast, 2024). However, high initial uptake does not automatically translate into sustained account use. Numerous empirical studies report mixed findings even for short-term usage rates (Dupas et al., 2018; Dupas and Robinson, 2013; Prina, 2015; Schaner, 2017), and institutions such as the Center for Financial Inclusion have raised concerns about high dormancy rates, particularly in South Asia and SSA (Demirgüç-Kunt et al., 2018; Rhyne and Kelly, 2018). This suggests that external barriers are only part of the problem: interventions that tackle them may have limited effects when internal, psychological constraints are present. In particular, recent theory highlights how low aspirations and

hope can generate behavioral poverty traps (Dalton et al., 2016; Genicot and Ray, 2017; Lybbert and Wydick, 2018; Ray, 2006). Building on these insights, a growing experimental literature tests whether raising aspirations and hope can improve economic outcomes (Beaman et al., 2012; Bernard et al., 2023; McKenzie et al., 2022).

Against this backdrop, we present results from a randomized controlled trial (RCT) in rural Ghana that tests two behavioral interventions—a video and a "savings calendar"—designed to encourage smallholder cocoa farmers to save more for old age. Our video focuses on positive testimonies from cocoa farmers who saved toward old age and realized their aspirations. Our savings calendar is an annual wall calendar that displays prominent motifs suggestive of cocoa farming and savings accumulation, in which those who received it were asked to write in the blank spaces of the calendar: (i) one long-term non-monetary savings goal for old age (e.g., owning a house) and (ii) a monthly savings amount toward achieving the stated long-term goal.

The video seeks to raise aspirations and hope regarding financial well-being later in life, while the savings calendar is intended to make the need to save for old age and farmers' specific savings goals more salient, aiming to sustain any effects of the video over time. We examine the impact of these treatments on smallholders' aspirations and hope, as well as on their (pension) savings.

Our first intervention—the video—is grounded in the literature on aspirations and hope. The literature on aspirations originates in the field of positive psychology, which focuses on factors that enhance human well-being (Seligman and Csikszentmihalyi, 2000). In particular, our video intervention draws on Ray's (2006) model, which introduces the aspirations window, i.e., the set of people one perceives as similar to oneself who are used as reference points when forming aspirations. Ray argues that future-oriented behavior is shaped by the aspirations gap: the difference between the living standard a person aspires to (defined by their aspirations window) and their current standard. According to him, people minimize a weighted average of this gap and the cost of investment needed to close it. A key result from Ray's analysis is that very small or very large aspiration gaps can lead to minimal investment, a phenomenon Ray calls aspirations failure. In short, aspirations lead to tangible changes only when the window is open but not too wide, i.e., when relevant role models exist but do not seem too far removed, as with the cocoa farmers in our sample who watched video testimonies from peers who had saved

successfully.

Building on Ray (2006), Dalton et al. (2016) argue that persistent aspirations failure can result in a self-perpetuating cycle of low aspirations and low living standards; interventions that raise aspirations, such as our video, may therefore be required to break the cycle. In agriculture, higher aspirations—often fostered by targeted interventions—have been found to be linked to greater technology adoption and investment and, in some studies, higher short-term savings (Bernard et al., 2023; Nandi and Nedumaran, 2021). Whether such aspiration gains translate into saving for old age, however, has yet to be tested—a gap this study aims to fill.

The concept of aspirations is closely related to the concept of hope. In his seminal work, Snyder (2000) defines hope as "a positive motivational state that is based on an interactively derived sense of successful (a) agency (goal-directed energy) and (b) pathways (planning to meet goals)". In other words, hope consists of two distinct elements: (1) agency, i.e., the perceived ability to follow the pathways toward one's aspirations, and (2) pathways, which refer to one's perceived ability to plan different routes that can be taken to meet those aspirations. Building on Snyder (2000), Lybbert and Wydick (2018) coined the term aspirational hope, combining aspirations with Snyder's hope framework of agency and pathways. Accordingly, this study examines effects on agency and pathways in addition to aspirations, as well as on (pension) savings. In line with Snyder's (2000) conceptualization, and consistent with much of the theoretical literature (see also Martin, 2014), which views hope as directed toward a desired outcome, we expect the sub-indices of hope to be affected by the intervention only after individuals form their aspirations.

A comprehensive review of hope and economic outcomes in low- and middle-income settings (Lybbert and Wydick, 2022) finds links to asset accumulation but identifies no evidence on pension saving. Only one study—an RCT with low-income mothers in the United States—links higher hope to larger voluntary retirement contributions (White et al., 2020). Thus, whether fostering hope can boost old-age saving among the poor—including the rural poor, most importantly smallholders—in developing countries remains untested, a gap this paper addresses.

Our calendar intervention is informed by two strands of evidence. First, the literature on aspirations interventions suggests that while short-term aspiration boosts can persist

(e.g., Orkin et al., 2023), their magnitude may taper (Bernard et al., 2023). Second, the savings literature underscores the importance of making savings goals salient. For example, Karlan et al. (2016) show that reminders can increase savings by making savings goals more salient, serving as a nudge, and helping individuals stay committed to their plans and take action toward their goals. Similarly, Soman and Zhao (2011) find that visual reminders that make savings goals more salient improve goal attainment.

Taken together, these insights led us to adopt a calendar that, by making the savings goals visible every day, keeps the video-induced aspirations salient, aiming to sustain the effects of the video intervention over the long term and ultimately improve (pension) savings.

We estimate the effect of our interventions and the hypotheses listed above using a sample of smallholder cocoa farmers in the Fanteakwa and Suhum cocoa districts in Ghana's Eastern Region. This focus on Ghanaian smallholder cocoa farmers is motivated by the following considerations. First, their savings constraints are well documented: like other smallholders, smallholder cocoa farmers face seasonal income spikes, long lean periods, and high vulnerability to shocks Kos and Lensink (2023), leaving few able to build reserves for old-age security. Second, they embody a region-wide demographic pattern. Across SSA the farm workforce is greying while the general population is predominantly young: the average farmer is now close to 60, whereas roughly 60 percent of Africans are under 24. Third, women provide a large share of farm labor, yet lack secure land rights, credit, and decision-making power (Nandi and Nedumaran, 2021). Given that aspiration failure, presumably more common among older and female farmers, can trap households in low-investment equilibria, these stylized facts underscore the importance of studying our particular sample of individuals and motivate our heterogeneity analysis described below.

To implement our study, we collaborated with People's Pension Trust (PPT), a licensed pension administrator for both low-income formal workers and informal workers in Ghana. We randomly assigned 956 cocoa farmers, all of whom had existing pension savings accounts with PPT (i.e., where there was less of a risk of binding external constraints to saving for old age), to two treatment groups (one combining the video with a regular "placebo" calendar, the other combining the video with the savings calendar) and a control group (that only received the placebo calendar and did not see the video),

stratified at the community level. The calendar intervention was implemented after the baseline, which included the video intervention at its end for those assigned to either of the two treatment groups, and after a subsequent remeasurement of aspirations (midline). This implied that respondents in the video plus savings calendar treatment group were the only ones presented with the savings calendar, and that the video plus placebo calendar treatment group and the control group both received the placebo calendar.

Our main outcome variables are survey-based measures of aspirations and hope, as well as (pension) savings. We measured aspirations at three points: at baseline (before the video intervention), immediately after the video intervention (midline), and 12 months after the baseline (endline). Hope, i.e., agency and pathways, was measured at baseline and again at endline. Note that when discussing our results below, we refer to midline and endline outcomes as short-term and long-term, respectively. Besides estimating the average impact of our interventions on aspirations, agency and pathways, we also conduct heterogeneity analysis to examine how these effects vary for two demographic variables, i.e., gender and initial savings level. We discuss each in turn.

First, for gender, women cocoa farmers may have fewer successful female role models, limiting their initial capacity to aspire. Moreover, women in agrarian economies frequently lack social and economic power (Ashraf, 2009; Beaman et al., 2012; Duflo, 2012), which can limit the effectiveness of interventions aimed at enhancing aspirations. Women's restricted access to resources and weaker intrahousehold bargaining power further curb women's agency.

Second, we examine heterogeneity by initial savings levels. This investigation is informed by related literature (Gehrke et al., 2023), which finds that treatment effects on educational aspirations are higher for those who already received better grades at baseline.

We find a significant positive effect of our video on aspirations in the short-term. While there are no significant long-term average treatment effects of the two treatments on aspirations, we find heterogeneity along two dimensions. First, the marginal effect on aspirations of the video in combination with the savings calendar is lower for females; though, overall, females are not worse off as a result of the treatment compared to women who did not receive the treatment. Second, higher initial levels of total savings predict higher marginal effects on aspirations of the video in combination with either the placebo

or the savings calendar.

We also find a positive long-run average treatment effect of the video in combination with the placebo calendar on farmers' hope, driven by an increase in agency. Even though the corresponding average treatment effect for the video in combination with the savings calendar is not statistically significant, we cannot reject the null hypothesis that it is equal to the treatment effect of the video in combination with the placebo calendar. Mirroring the result for aspirations, we find that the long-run marginal effects of the video in combination with the savings calendar on agency and pathways—and thus also on hope—are significantly smaller for females. However, in contrast to the results for aspirations, we find that females are overall worse off as a result of the treatment in terms of agency and hope (but not pathways) than had they not received the treatment. This result is intuitive: While the video in combination with the savings calendar treatment does not influence women's pathways (i.e., how to translate their aspirations into reality), their lack of opportunities to follow those pathways negatively affects their sense of agency, thereby lowering their overall hope.

Importantly, the video in combination with the savings calendar significantly increases the fraction of farmers saving for old age in the long run, and the treatment effect of the video with savings calendar on the average amount of pension savings is significantly higher among those smallholders who had higher initial levels of total savings. This result mirrors our finding about higher initial levels of total savings predicting higher marginal effects on aspirations and is once again reminiscent of the findings of Gehrke et al. (2023) regarding educational outcomes.

Our study contributes to the growing literature on how psychological and behavioral interventions can promote long-term savings among the rural poor, and more specifically the role of aspiration-raising interventions. While existing papers focus primarily on the effects of aspiration-enhancing interventions on business and educational outcomes (Beaman et al., 2012; Bernard et al., 2019, 2023; Galiani et al., 2021; McKenzie et al., 2022), to the best of our knowledge, ours is the first to measure their effects on pension savings. Hence, we contribute to the topical research and policy discourse on old-age welfare of the poor in developing countries.

Several studies suggest that exposure to socially relatable role models can enhance aspirations and downstream outcomes. Bandura (1977) describes this as the power of

vicarious experience: seeing similar others succeed increases aspirations. Lybbert and Wydick (2017) showed that a short documentary on successful women entrepreneurs, combined with a goal-setting magnet and a workshop on aspirations, agency, and pathways, raised aspirations and modestly increased agency among female community bank members in Mexico after a month. Rojas Valdes et al. (2022) developed a 25-minute video on successful borrowers, supplemented by periodic follow-up messages, and observed positive short-term (1 month) effects and long-term (1 year) effects on aspirations, agency and pathways. Cecchi et al. (2022) implemented a similar intervention in Bolivia and found increased production aspirations among dairy farmers. Beaman et al. (2012) found that exposure to female leadership positions in village councils in India, created via a gender quota system, substantially increased educational aspirations and outcomes of young women. Bernard et al. (2019) invited Ethiopian households to watch a one-hour documentary about local individuals who improved their lives, resulting in higher educational aspirations and investments. Riley (2024) showed a 2-hour film about a female role model who succeeded in becoming a chess master through hard work and perseverance to students in Uganda and found improved exam results among viewers compared to a group which watched a placebo film.

Other studies take as a starting point that aspirations are socially formed, even when not explicitly focused on role models. Garcia et al. (2020) found that a microfinance group lending program in Sierra Leone increased life and future economic aspirations. Chiapa et al. (2012) show that an anti-poverty program in Mexico enhanced parents' educational aspirations for their children. Macours and Vakis (2014) provided evidence that cash transfers in Nicaragua improved human capital, though aspirations were not directly measured.

# 2 Study setup

## 2.1 Study setting and partner organization

We focus on smallholder cocoa farmers in Ghana, a setting where economic vulnerability in later life is widespread and where pension savings remain limited. Smallholder agriculture dominates rural livelihoods across SSA: farms under two hectares constitute the bulk of producers and supply a substantial share of food (Lowder et al., 2016). In

Ghana, agriculture accounts for roughly one-fifth of GDP and around one-third of total employment, while cocoa is the country's most important cash crop and supports the livelihoods of about 800,000[1].Despite its economic significance, cocoa production is dominated by low-income smallholders who face persistent exposure to income volatility, food insecurity, and poverty. Structural changes in rural Ghana have further heightened old-age vulnerability. Out-migration of younger household members, urbanization, and rising educational aspirations have reduced the availability of intergenerational support, weakening the traditional family-based safety net (Ao et al., 2016; Bezu and Holden, 2014). At the same time, formal old-age protection remains limited: contributory pension schemes cover only a minority of workers, with most rural and informal-sector adults lacking access to retirement benefits. As a result, many smallholder farmers enter old age with little accumulated savings and few reliable sources of support. These conditions underline the policy relevance of interventions aimed at strengthening pension savings among rural households and improving their financial security in later life.

To implement our study, we collaborated with People's Pension Trust (PPT), a licensed pension administrator for both low-income formal workers and informal workers in Ghana. Notably, when PPT expanded its voluntary contributory pension scheme coverage, providing farmers in rural areas – including smallholders in the Fanteakwa and Suhum cocoa districts in Ghana's Eastern Region – the opportunity to save toward old age, it observed very low usage of its pension savings product among farmers registered with PPT. Our sample consists of cocoa farmers in the Fanteakwa and Suhum cocoa districts who have already opened PPT accounts. This makes the sample especially well-suited to our research question, as these farmers do not face binding external constraints to saving formally for their old age, making our treatments that target internal constraints especially relevant for this group. At the same time, since the challenges confronting Ghanaian cocoa farmers in these respects are broadly similar to those of other smallholders in SSA, the setting also offers useful insights that are likely to extend beyond this particular context.

[1]Source: World Development Indicators by World Bank (2023); and Ministry of Food and Agriculture (2021).

## 2.2 Interventions and experimental design

Participants were randomly assigned to one of three groups: a control group which received a placebo calendar; a treatment group that received a video plus placebo calendar (*Video + Plac. Cal.*) treatment; and a treatment group that received a video plus savings calendar (*Video + Sav. Cal.*) treatment.

**Control group.** The control group received a regular calendar as placebo, which was an annual wall calendar with a decorative picture of a rose (see Appendix Figure A.1 for the design). Control group participants were given this calendar to treat them at par with treatment group participants, to prevent frictions that might have affected outcome measures.

**Treatment group: *Video + Plac. Cal.*** Farmers in this group were given the same placebo calendar as the control group (to prevent frictions, but also to rule out that the savings calendar in the Video + Sav. Cal. treatment group simply served as a reminder of the video intervention) and shown a video in addition. The video focused on positive testimonies from farmers who saved toward old age and realized their aspirations. The content emphasized how regular, small savings can improve old-age well-being for cocoa farmers (see screenshot from the video in Appendix Figure A.2), without mentioning how to save (i.e., we did not promote any specific ways to save for old age or mention any specific savings product). Those featured in the video (i.e., the actors) are actual cocoa farmers, from similar communities as the ones of the farmers in our sample, though they belonged to a different cocoa district. The video had a running time of 5 minutes 21 seconds, out of which men share their experiences/opinions for 2 mins 41 seconds and women for 1 min 15 seconds, and of which 22 seconds were devoted to a young man asking a question. Farmers were semi-scripted. The voices were recorded in Twi, which is the main language spoken in the area, and the video was accompanied by English subtitles. The video was stored on the tablet of each enumerator and was shown to each participant individually and privately to avoid external influences such as community effects. Participants who had watched the video and completed the midline were asked to immediately leave the location in order to prevent discussions of the video with other participants (see also Section 4.3 for the full timeline of the intervention).

**Treatment group: *Video + Sav.Cal.*** Individuals in this group were shown the video described above and received a savings calendar. This calendar was also an annual wall calendar, like the placebo calendar, but differed from the latter in several important ways (see Appendix Figure A.1 for the design). Instead of the picture of a rose as in the placebo calendar, the savings calendar depicted a young cocoa tree, relating it to the principal income-generating activity of cocoa farming for our participants, and displayed other prominent motifs suggestive of progressive savings.

The design also featured two empty white rectangular spaces at the base of the tree where the owner of the calendar could indicate (i) a concrete, non-monetary, long-term savings goal for old age (e.g., own a house, being able to finance educational attainment of children), and (ii) a monthly financial savings target toward achieving the long-term goal. Participants in this treatment group were asked, after watching the video, to pencil (i) and (ii) into the two blank spaces provided in the savings calendar. These two self-inscriptions were meant to increase the salience of the need to save for old age and of farmers' specific savings goals for old age; hence they were encouraged to hang these calendars where they would see them every day.

## 2.3 Timeline and data collection procedure

The timeline of our field activities is depicted in Figure 1. The interventions and baseline data collection were carried out from from 19 September 2019 to 1 November 2019. Farmers were asked to report to a community center, a community school or a church building and were then interviewed individually, to ensure privacy, by trained enumerators using a structured questionnaire. To limit enumerator influence on farmers' responses, questions regarding aspirations and hope (i.e., agency and pathways) were pre-recorded and played to each participant. Those in either of the two treatment groups were shown the video on a tablet next. All participants were asked to respond to the aspirations statement for a second time (hereafter, the midline survey, reflecting its timing). The endline took place between 13 October 2020 and 23 November 2020.[2] Note that data collection was done by the Social Science and Statistics Unit of the Cocoa Research Institute of Ghana

[2]The endline survey was initially planned to be administered six months after the baseline (i.e., March to April 2020) but was disrupted by nationwide restrictions in response to the Covid-19 pandemic. Thus, we conducted the endline survey from October to November 2020, once Covid-19 restrictions were relaxed sufficiently.

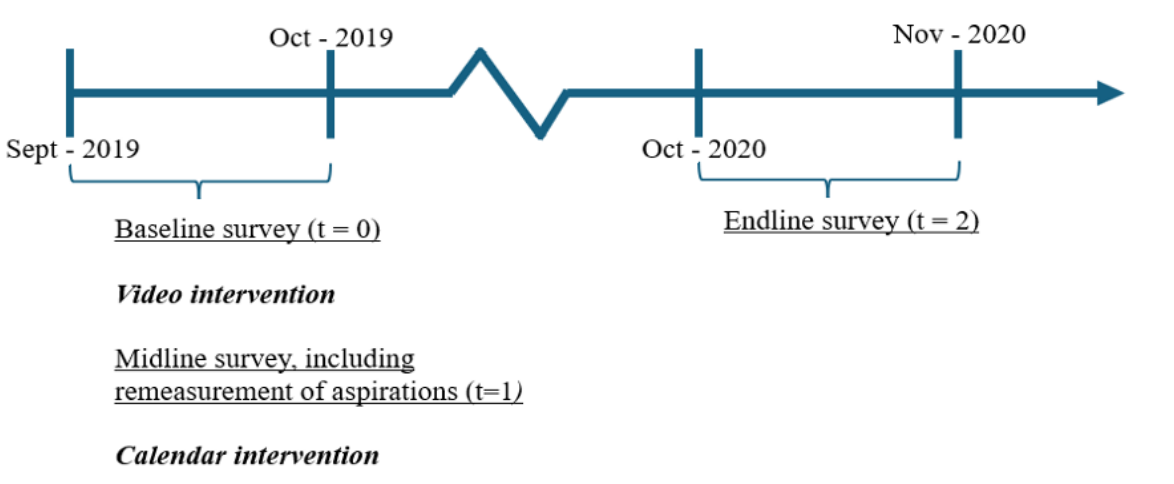


Figure 1: Timeline of the study

(CRIG) with CRIG, a subsidiary of the Ghana Cocoa Board (COCOBOD). Thus, our enumerators were not affiliated with PPT.

# 3 Methodology

Our general approach to study treatment effects is to use analysis of covariance (ANCOVA).[3] For Aspirations, which was measured thrice (at baseline, at midline and at endline; see Section 3.4), we estimate the immediate average treatment effect using Aspirations measured at midline as described below in Section 4.1, as well as the long-term average treatment effect using Aspirations measured at endline as described in Section 4.2. For the Hope, Agency and Pathways indices, as well as the savings outcomes, we assess long-term average treatment effects, as described in Section 4.2. In addition, for all psychological and savings outcomes, we assess heterogeneity in long-term treatment effects using selected baseline characteristics, as described in Section 4.3. The motivation for the choice of these baseline characteristics is elaborated on in the literature review, Section 2.3. In all cases, we first regress our outcome variables on their baseline values and the treatment dummies. We then repeat the analysis controlling for variables selected on the basis of our balance tests and attrition analysis (see Section 4.4 for these analyses). Note that we also select controls based on a double lasso methodology. This

[3]We use ANCOVA for all outcome variables except for PPT savings, for which we do not have baseline values. PPT savings are analyzed using simple post-treatment estimates, for which the equations estimated are identical to (3)-(6), except that they do not include the baseline value of the dependent variable as a regressor.

does not affect the results (See Appendix F).

## 3.1 Immediate effect of video intervention

To estimate the immediate average treatment effect of the video on Aspirations, we use the following regression:

$$Y_{i,t=1} = \alpha + \beta_1 \, Video_i + \beta_2 Y_{i,t=0} + \gamma X_{i,t=0} + \varepsilon_i, \tag{1}$$

where subscript $i$ represents the $i$-th individual and $t = 0$ represents a baseline measurement; $t = 1$ represents a midline measurement; $Y$ represents Aspirations; and the dummy variable *Video* is an indicator for having watched the aspirations video (= 1 if Yes), i.e., for belonging to either of the two treatment groups. $X_{i,t=0}$ are the controls variables.

## 3.2 Long-term effect of two treatments

Proceeding in an analogous manner, we assess the long-term effect of the interventions on the psychological and savings outcomes by first estimating a regression model only with treatment dummies and the baseline value of the dependent variable with control variables:

$$Y_{i,t=2} = \alpha + \beta_1 (Video + Plac.\ Cal.)_i + \beta_2 (Video + Sav.\ Cal.)_i + \beta_3 Y_{i,t=0} + \gamma X_{i,t=0} + e_i, \tag{2}$$

where subscript $t = 2$ represents an endline measurement. All other notation remains unchanged.

To separately identify the marginal effect of the savings calendar in the long term, we create a dummy variable for watching the video (1 for those who watched and 0 for those who did not watch). We label this variable as $AnyVid_i$ to indicate that it captures having watched the video as part of either of the two treatment groups. Proceeding as before, we estimate the following two models, where the coefficient on $AnyVid_i$ is the average marginal effect of having watched the video as part of *either* treatment group and the coefficient on $Video + Sav.\ Cal.$ is the marginal effect of the savings calendar over and above that effect, both in the long run:

$$Y_{i,t=2} = \alpha + \beta_1 AnyVid_i + \beta_2(Video + Sav.\ Cal.)_i + \beta_3 Y_{i,t=0} + \gamma X_{i,t=0} + e_i. \quad (3)$$

### 3.3 Long-term heterogeneous treatment effects

Following the arguments presented in the introduction, we do not expect treatment effects to be the same across certain characteristics such as gender, age, and initial levels of savings. Hence, we conduct heterogeneity analyses of treatment effects with respect to these key variables as outlined below. Specifically, we estimate the following equation, considering interactions between treatments and the selected demographic or socio-economic variables:

$$\begin{aligned} Y_{i2} = \alpha &+ \beta_1(Video + \text{Plac. Cal.})_i + \beta_2(Video + \text{Sav. Cal.})_i \\ &+ \beta_3(Video + \text{Plac. Cal.})_i \times D_{i,t=0} \\ &+ \beta_4(Video + \text{Sav. Cal.})_i \times D_{i,t=0} \\ &+ \beta_5 D_{i,t=0} + \beta_6 Y_{i,t=0} + \gamma X_{i,t=0} + e_i. \end{aligned} \quad (4)$$

where $D$ is the baseline ($t = 0$) value of the variable of interest, i.e., gender and total savings, all measured at the individual level ($i$), and $X_{i0}$ is the union of $D_{i0}$ and our earlier set of baseline controls. All other notation remains the same.

## 4 Data

### 4.1 Sampling frame, randomization and determination of planned sample

The sampling frame for our study consisted of all cocoa farmers belonging to the Fanteakwa and Suhum cocoa farming cooperatives in the Eastern Region of Ghana who were registered with PPT, i.e., who had opened pension savings accounts up until August 2019. These farmers belonged to 56 communities. As a maximum of 22 individuals per community was (randomly) chosen, the number of eligible individuals per community varied widely, based on the size of a community, ranging from 1 to 22. This led to a planned sample size of 956 farmers. Stratification was then done at the community level, with eligible individuals being randomly assigned to the control and two treatment groups.

## 4.2 Communication prior to interventions

Each cooperative operated with an overall chairperson and executive members overseeing local chairpersons at the community level. To maximize the size of our realized sample from the sampling frame described above, we contacted the overall chairperson of each of the two cooperatives, and in subsequent visits to their offices briefed them and the executive members about the survey we wanted to undertake and the objectives of the project in general terms. The next step was to schedule the surveys and interventions in the field. To do so, we contacted the local community chairpersons by phone, when possible, about two weeks before the start of the surveys (by which time they had been made aware of our project by the overall chairperson of their cooperative). In some cases, where phone contact was unsuccessful, we had to travel to the community to make arrangements. In the majority of cases, to ensure maximum participation, the surveys and interventions were scheduled on either the local cooperative meeting day (during which voluntary savings were collected by a PPT representative) or on the local "Taboo days".[4] In most communities, these two days coincided. We asked the local chairpersons to inform, on our behalf, the farmers we had included in our sample that there would be a survey (executed by a CRIG representative, as noted in Section 3.4; CRIG being an organization with which the farmers had strong prior ties). Contacting the chairpersons two weeks in advance provided them ample opportunity to inform those selected in time, and also possibly to remind them of the survey/study during the following week's meeting. We called or visited the local chairpersons again within a week after the first call or visit, and visited them as a reminder when their communities were next in line to be surveyed.

On the day of the meeting and before the survey took place, the project objective was explained to each farmer in general terms. We also gave them an estimate of the time it would take to complete the survey and informed them that they would receive a cutlass (a commonly used tool in cocoa farming) as an in-kind show-up fee. The farmers were then asked to sign a consent form if they wished to participate in the survey and interventions.

[4] "Taboo days" are days on which farmers do not farm, to provide rest for both the land and themselves.

Table 1: Realized sample across survey rounds and treatment groups

| Survey Round | Control | Vid. + Plac. Cal. | Vid. + Sav. Cal. | Sum |
|---|---|---|---|---|
| Baseline | 235 | 240 | 248 | 723 |
| Endline | 197 | 208 | 215 | 620 |

*Notes:* The first and second rows of the table show the group assignment of the cocoa farmers who could be interviewed for the baseline and endline surveys, respectively. See text for details.

## 4.3 Realized sample

A total of 723 and 620 cocoa farmers out of the planned sample were interviewed for the baseline and endline, respectively (see last column of Table 1). The first row of Table 1 shows the group assignments of the 723 cocoa farmers who could be interviewed for the baseline: 235 farmers were assigned to the control group, 240 farmers to the Vid.+ Plac. Cal. treatment group, and 248 to the Vid.+ Sav. Cal. treatment group. For the endline, 197 of those in the control group, 208 of those in the Vid.+ Plac. Cal. treatment group, and 215 of those in the Vid.+ Sav. Cal. treatment group could be successfully re-interviewed.

## 4.4 Outcome variables

**Psychological outcome variables: Aspirations and Hope.** Our psychological outcome variables are Aspirations regarding financial well-being in old age and three hope-related indices, i.e., Hope, and its two sub-indices Agency and Pathways. To construct these four variables, we used nine statements (see Table 2) based on existing work, one concerning Aspirations, and four each concerning Agency and Pathways. Specifically, the Aspirations statement was adapted from Lybbert and Wydick (2018) to the context of old age. The Hope index consisted of four statements that measured Agency and four statements that measured Pathways, which were retained in their original wording from the Adult Hope Scale (see Snyder, 1994).

We asked subjects to indicate their level of agreement with each of the statements on a six-point Likert scale ranging from 0 (strongly disagree) to 5 (strongly agree). The resulting Aspirations scores were standardized by subtracting the mean for the control group at baseline and dividing by the standard deviation of the control group at baseline

Table 2: Statements used to construct the psychological outcome variables

| **Outcome variable** | **Statement** |
|---|---|
| *Aspirations* | 1: It is important to set financial goals when planning for old age / the future. |
| *Agency* | 1: "You've been pretty successful in life".<br>2: "You energetically pursue your goals".<br>3: "Your past experiences have prepared you well for your future".<br>4: "You meet the goals that you set for yourself". |
| *Pathways* | 1: "You can think of many ways to get out of a jam".<br>2: "There are lots of ways around any problem".<br>3: "You can think of many ways to get the things in life that are important to you".<br>4: "Even when others get discouraged, you know you can find a way to solve the problem". |

*Notes:* The table lists the statements used to construct our main psychological outcome variables: *Aspirations* and *Hope* (i.e., *Agency* and *Pathways*, see text for further details). The Aspirations statement (Statement 1) was adapted from Lybbert and Wydick (2018) to the context of old age. The Hope index consisted of four statements that measured Agency and four statements that measured Pathways, which were retained in their original wording from the Adult Hope Scale (see Snyder, 1994).

(Kling et al., 2007; Lybbert and Wydick, 2018), as shown below (equation 5):

$$\frac{e - \bar{e}_{bc}}{\sigma_{\bar{e}_{bc}}}, \tag{5}$$

where $e$ is the Likert scale level of the response, and $\bar{e}_{bc}$ and $\sigma_{\bar{e}_{bc}}$ are respectively the mean and standard deviation of the score for the control group measured at the baseline.

The approach was nearly identical for Agency, Pathways, and Hope, except that the average ($\nu$ below) of the responses to the corresponding questions was computed first (i.e., by dividing the sum of the responses to the corresponding questions by the number of corresponding questions), and the mean and standard deviation of $\nu$ were then used for standardization, as shown below (see equation 6):

$$\frac{\nu - \bar{\nu}_{bc}}{\sigma_{\bar{\nu}_{bc}}}, \tag{6}$$

where $\nu = \left(\frac{\sum e}{number\ of\ questions}\right)$ is the average of the Likert scale levels of the responses $e$

for all corresponding questions, and $\bar{\nu}_{bc}$ and $\sigma_{\bar{\nu}_{bc}}$ are respectively the mean and standard deviations of $\nu$ for the control group measured at the baseline.

**Outcome variables related to pension savings.** In addition to the psychological outcome measures described above, we also considered the impact of the interventions on relevant savings outcomes, specifically, the fraction of farmers who report saving for old age (Saving for old age) as well as the amount of savings farmers store in their PPT accounts, measured in Ghana Cedis (PPT savings).

## 4.5 Explanatory variables

Besides the outcome variables, we collected additional variables via the baseline and endline surveys. These variables include demographics, socio-economic status as well as shocks faced by households (see Table 3). Based on our balance test results and attrition analysis (see Section 4.6), we include several such variables as controls.

## 4.6 Descriptive statistics, balance and attrition

Table 3 provides summary statistics of our sample at baseline. Consistent with the general demographics of cocoa farmers in Ghana, the study respondents are predominantly male (80.4%) with only 19.6% being female. Their mean age is 55.8 years with an average of 9.3 years of formal education. Respondents had an average household size of almost 6 and an average of 9.3 acres self-owned cocoa farms. They also had high levels of aspirations (a non-standardized average score of 3.58 out of a maximum of 5) and hope (a non-standardized average score of 4.05 out of a maximum of 5) to begin with.

The balancing tests, as shown in Appendix Table B.4, confirm that the randomization was largely successful, with only a few variables exhibiting significant differences in means across the three groups. It is important to note that the realized baseline sample used for the balancing tests is smaller than the planned sample that was randomized into the control and treatment groups (see Table 1). Nevertheless, the results of the balancing tests indicate that the sample remains well-balanced.

We present results with and without controlling for these unbalanced variables in all subsequent analysis. These variables are female, age, migration status, farm equipment, social capital, financial difficulty and natural/biological disaster. Note that out of the 69

Table 3: Descriptive statistics

| Category | Variables | N | Mean | SD | Min | Max |
|---|---|---|---|---|---|---|
| **Main Outcomes** | | | | | | |
| | Aspirations (Bsl.) | 723 | 0.040 | 0.951 | -7.659 | 0.660 |
| | Agency (Bsl.) | 723 | 0.090 | 1.018 | -3.521 | 1.590 |
| | Pathways (Bsl.) | 723 | 0.028 | 1.076 | -5.981 | 1.337 |
| **Demographics** | | | | | | |
| | Female | 723 | 0.196 | 0.398 | 0 | 1 |
| | Age | 721 | 55.854 | 12.547 | 24 | 97 |
| | Household size | 723 | 5.660 | 2.460 | 1 | 23 |
| | Migration status | 722 | 0.309 | 0.462 | 0 | 1 |
| | Speaks Twi | 723 | 0.819 | 0.385 | 0 | 1 |
| | Education | 705 | 9.275 | 2.833 | 0 | 20 |
| | Financial literacy | 723 | 0.642 | 0.480 | 0 | 1 |
| **Income and Savings** | | | | | | |
| | Income | 662 | 11.261 | 71.293 | 0 | 1,806 |
| | Total farm size | 717 | 9.333 | 19.376 | 0 | 252 |
| | Total savings (Bsl.) | 423 | 1.310 | 2.833 | 0 | 31.2 |
| | Saving for old age | 723 | 0.570 | 0.495 | 0 | 1 |
| | Salience of old age | 723 | 0.678 | 0.468 | 0 | 1 |
| **Other Assets** | | | | | | |
| | House size | 723 | 4.113 | 2.407 | 1 | 15 |
| | Elec. and electr. assets | 723 | 3.195 | 1.572 | 0 | 8 |
| | Vehicles | 723 | 0.451 | 0.646 | 0 | 3 |
| | Farm equipment | 723 | 0.845 | 0.662 | 0 | 2 |
| **Social Standing** | | | | | | |
| | Leadership status | 723 | 0.159 | 0.366 | 0 | 1 |
| | Social capital | 723 | 2.183 | 1.184 | 0 | 4 |
| **Shocks** | | | | | | |
| | Financial difficulty | 723 | 0.172 | 0.377 | 0 | 1 |
| | Natural/biological disaster | 723 | 0.351 | 0.478 | 0 | 1 |

*Notes:* Female = female respondent; Age = respondent's age; Household size = number of people living in the same household (i.e., sharing a dwelling and housekeeping under one recognized head). Migration status is a dummy equal to 1 for first-generation migrants (non-natives of the current community) and 0 otherwise. Speaks Twi is a dummy equal to 1 if the respondent speaks Twi and 0 otherwise. Education is years of formal schooling. Financial literacy is based on four questions (one math, one inflation, two interest rate); respondents answering at least three correctly are coded 1, and 0 otherwise. Income is calculated as annual cocoa yield multiplied by GH¢ 475 per 64 kg bag (the 2018/2019 government producer price). Total farm size is total acreage owned; respondents may also operate additional plots under tenure arrangements such as Abunu, Abusa, and Abunan. Savings for old age is a dummy equal to 1 if the respondent saves for old age. Salience of old age equals 1 if the respondent often thinks about old age. House size is the number of rooms. Elec. and electr. assets count categories owned (e.g., cellphone, freezer, fridge, laptop, radio, satellite dish, tablet, television). Vehicles count transport asset categories (bicycle, car, motorbike, tricycle, truck). Farm equipment counts categories owned, mainly knapsack sprayer and mist blower. Leadership status equals 1 if the respondent holds a leadership role in the community or farmer group. Social capital counts categories of important community contacts (e.g., chief, spiritual leader, farmer group leader, certification manager). Financial difficulty equals 1 if the respondent sold assets due to debt distress. Natural/biological disaster equals 1 if the respondent experienced flooding, pests, or disease in the past year before the intervention.

comparisons in the balance table, only three are significant at 5% and one at 1%. Since at these levels, approximately 3.5 and 0.69, respectively, of these differences are expected to be significant simply by chance, we are not overly concerned about our sample being unbalanced (despite 10 differences being significant at 10%, which exceeds the expected number of 6.9).

As mentioned earlier, 724 farmers participated in the baseline survey, but the endline involved 622 farmers. Thus, some level of attrition was observed. To confirm that attrition was random, and since there was no attrition at midline, we ran two regressions as follows:

$$Attendance_{i,t=2} = \alpha + \beta_1 \left(\text{Video} + \text{Plac. Cal.}\right)_i + \beta_2 \left(\text{Video} + \text{Sav. Cal.}\right)_i + \gamma X_{i,t=0} + e_i, \quad (7)$$

where Attendance is an indicator variable for whether respondents were interviewed at endline ($t = 2$ for Yes); $X$ is a vector of outcome and explanatory variables (note that these are not the same $X_{i,t=0}$ used in our main analysis, which are control variables selected on the basis of the balance tests and the attrition analysis, described above), and the remaining notation is as before. The attrition tests (see Appendix B) indicate just one variable (Salience of old age) to be a negative predictor of attrition, which we control for in all subsequent analysis that includes the unbalanced variables as controls.

# 5 Empirical results

## 5.1 Short-term effect of video intervention on aspirations

Recall that we recorded participants' Aspirations as part of the baseline survey. For those in the control group, we simply recorded their Aspirations a second time right after the baseline, while for the treatment groups, we recorded their Aspirations once more immediately after they watched the video. We find a statistically significant (at 5%) positive impact of the video intervention on Aspirations. Column 1 of Table 4 shows that the video intervention immediately raises the aspiration score by 0.15 standard deviations. When we add controls, the treatment effect of 0.12 standard deviations remains significant at 10%.

These results are in line with previous studies that reported positive short-term to medium-term impacts of similar interventions (e.g., Bernard et al., 2019; Ross et al., 2021;

Table 4: Short-term impact of video intervention on aspirations

| VARIABLES | (1)<br>Aspirations (Mdl.) | (2)<br>Aspirations (Mdl.) |
|---|---|---|
| Video | 0.149** | 0.120* |
| | (0.073) | (0.070) |
| Aspirations (Bsl.) | 0.513*** | 0.518*** |
| | (0.074) | (0.075) |
| Constant | 0.044 | 0.382* |
| | (0.061) | (0.195) |
| Baseline controls | No | Yes |
| Observations | 618 | 618 |
| R-squared | 0.262 | 0.282 |

*Notes:* The table reports the results of an ANCOVA specification to assess the video treatment effect on aspirations. Columns 1 and 2 present estimates without and with control variables respectively. Controls are variables that were unbalanced at the baseline. These are Female (equals "1" when the subject is female; "0" otherwise); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Full table showing coefficients of controls is presented in Appendix 3C. Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Bhan, 2021; Orkin et al., 2023; Lybbert and Wydick, 2018; Rojas Valdes et al., 2022) that were intended to enhance psychological attributes such as aspirations. For instance, Rojas Valdes et al. (2022) found a significant positive short-term (one month after the intervention) effect of a video intervention on the aspirations of female entrepreneurs.

## 5.2 Long-term treatment effects on aspirations, hope and its components, agency and pathways

**Aspirations.** Does the significant immediate increase in measured Aspirations uncovered in Section 5.1 persist in the longer run? Recall from Section 2.3 that after recording the immediate impact of the video intervention, we followed up with the savings calendar intervention, which was designed to sustain the increase in Aspirations (and Hope) over the longer term by increasing the salience of the need to save for old age and of

Table 5: Long-run treatment impact of Video + Plac. Cal. and Video + Sav. Cal. on psychological outcomes

| VARIABLES | (1) Aspirations | (2) Hope | (3) Agency | (4) Pathways |
|---|---|---|---|---|
| *Video + Plac. Cal.* | 0.017 | 0.142** | 0.147** | 0.064 |
| | (0.082) | (0.072) | (0.075) | (0.068) |
| *Video + Sav. Cal.* | 0.071 | 0.058 | 0.042 | 0.057 |
| | (0.084) | (0.068) | (0.073) | (0.070) |
| Aspirations (Bsl.) | 0.012 | | | |
| | (0.039) | | | |
| Hope (Bsl.) | | 0.132*** | | |
| | | (0.030) | | |
| Agency (Bsl.) | | | 0.147*** | |
| | | | (0.030) | |
| Pathways (Bsl.) | | | | 0.033 |
| | | | | (0.024) |
| Constant | -0.844*** | -0.335** | -0.181 | -0.407*** |
| | (0.196) | (0.164) | (0.177) | (0.149) |
| Baseline controls | Yes | Yes | Yes | Yes |
| Observations | 618 | 618 | 618 | 618 |
| R-squared | 0.035 | 0.085 | 0.088 | 0.048 |
| Test of differences in treatment coefficients | XXX | F(1,606)=1.57 Prob>F=0.211 | F(1,606)=2.46 Prob>F=0.117 | F(1,606)=0.01 Prob>F=0.925 |

*Notes:* The table reports the results of an ANCOVA specification to assess the heterogeneous treatment effect of gender (variable Female, which takes a value of 1 when the subject is female; 0 otherwise) on hope, agency and pathways. The two treatment variables are video plus placebo calendar (*Video + Plac. Cal.*) and video plus savings reminder calendar (*Video + Sav. Cal.*). The controls are Female (defined above); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Full table showing coefficients of controls is presented in Appendix C. Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

farmers' specific savings goals for old age. Recall as well that the video plus placebo calendar treatment group and the control group both received a placebo calendar. Table 5 presents ANCOVA estimates of the effects of the two treatments, the video with the placebo calendar (Video + Plac. Cal.) treatment and the video with the savings calendar (Video + Sav. Cal.) treatment, on long-term (i.e., measured one year after the interventions) Aspirations.

The results reveal no statistically significant long-term average treatment effect on subjects' aspirations. The significant immediate effect on Aspirations resulting from watching the video which was observed in Table 4 appears to have dissipated in the longer run. These findings align with a broader literature on aspiration-raising inter-

ventions. Bernard et al. (2019) similarly found that exposure to local role models in Ethiopia increased aspirations in the short term, but that these effects faded over time. Rojas Valdes et al. (2022) also observed that aspirations gains dissipated after one year. Comparable interventions by Cecchi et al. (2022) and Beaman et al. (2012) likewise found that exposure to relatable role models can enhance aspirations and goal-directed behavior, though sustaining these effects over time remains a challenge.

**Hope and its components, Agency and Pathways.** Unlike the results we find for the average long-term impact of our treatments on Aspirations, the results reveal a statistically significant positive long-run average treatment effect of the Video + Plac. Cal. treatment on subjects' Hope: For participants who received the Video + Plac. Cal. treatment, Hope increased by 0.142 standard deviations without and with control variables, respectively (see Column 2 of Table 5). This effect is driven by an increase in perceived agency: The Video + Plac. Cal. treatment increased Agency by and 0.147 standard deviations (see Column 3 of Table 5). Note that even though the corresponding average treatment effects for the Video + Sav. Cal. treatment are not statistically significant, testing whether the coefficients for the Video + Plac. Cal. treatment are equal to those for the Video + Sav. Cal. treatment in all cases where the former is significant but the latter is not revealed that we cannot reject the null hypothesis of equality.

Since these findings suggest that the video, rather than the calendar intervention, may be the primary driver of the observed increases in Hope and Agency, we estimate the separate marginal effects of having watched the video as part of either treatment group and of the savings calendar over and above that effect as explained in Section 3.2. We find that the coefficient on "AnyVid" (i.e., the marginal effect of having watched the video as part of either treatment group) is positive and significant and the coefficient of Video + Sav. Cal. is insignificant. This is further evidence that the video intervention positively enhanced subjects' Hope (via Agency) in the long term (see Table D.1 in Appendix D).

At the same time, we do not find any statistically significant long-run average treatment effects on Pathways. This is consistent with some previous research but contrasts with others. For instance, Lybbert and Wydick (2018) reported no treatment effect on pathways one month after their intervention, despite it including training on envisioning pathways out of poverty. In contrast, Rojas Valdes et al. (2022) found a significant positive effect on pathways 12 months after their video intervention. Appadurai (2007)

suggests that pathways may be particularly rigid for the poor, making them difficult to influence through one-shot interventions. In our case, the video intervention focused on positive testimonies from cocoa farmers who saved toward old age and realized their aspirations, emphasizing that regular, small savings can improve well-being in old age, but without offering concrete guidance on how to save or which products to use. Given that therefore neither our video nor our calendar intervention specifically targeted pathways for pension savings, the null results are not surprising.

## 5.3 Heterogeneous treatment effects on aspirations, hope and its components, agency and pathways

**Gender.** The introduction outlined our choice of variables for the heterogeneity analysis (gender, and initial level of savings). Table 6 (Column 1) presents ANCOVA estimates that show a statistically significantly lower effect of the Video + Sav. Cal. treatment on Aspirations for females. Specifically, the results show that the Video + Sav. Cal. treatment effect on aspirations is significantly lower than that of males by 0.461 standard deviations. Mirroring the result for Aspirations, we find that the long-run effects of the Video + Sav. Cal. treatment on Agency and Pathways – and thus also on Hope – are statistically significantly smaller for females (see Columns 2-4 in Table 6). More precisely, the results show that being female is associated with a reduction of the Video + Sav. Cal. treatment effects on Hope, Agency and Pathways by 0.451, 0.422 and 0.347 standard deviations, respectively. However, in contrast to the result for Aspirations, we find that females are also overall (statistically significantly) worse off as a result of the treatment in terms of Hope and Agency (but not Pathways) compared to not receiving the treatment, as shown by the Lincom estimates in Table 6.

This result is intuitive. Recall that the savings calendar intervention was designed to increase the salience of the need to save for old age and of specific savings goals, requiring farmers to pencil these goals into the empty fields of the calendar. We speculate that making these goals more salient by having participants write them down may have had an unintended negative impact on female participants: While participants in the Video + Plac. Cal. treatment, who had not written their goals, could more easily adjust them to better fit the changing external circumstances, the written goals in the Video + Sav. Cal. treatment may have been perceived as a commitment in the backdrop of these

Table 6: Heterogeneous treatment effect by gender

| VARIABLES | (1) Aspirations | (2) Hope | (3) Agency | (4) Pathways |
|---|---|---|---|---|
| *Video + Plac. Cal.* | 0.019 | 0.146* | 0.166** | 0.050 |
| | (0.089) | (0.080) | (0.084) | (0.075) |
| *Video + Sav. Cal.* | 0.154* | 0.139* | 0.116 | 0.122 |
| | (0.090) | (0.075) | (0.081) | (0.076) |
| Female | 0.224 | 0.318** | 0.346** | 0.175 |
| | (0.192) | (0.148) | (0.149) | (0.149) |
| Female × (Video + Plac. Cal.) | -0.111 | -0.119 | -0.181 | -0.010 |
| | (0.229) | (0.188) | (0.181) | (0.190) |
| Female × (Video + Sav. Cal.) | -0.461* | -0.451** | -0.422** | -0.347* |
| | (0.242) | (0.187) | (0.184) | (0.198) |
| Aspirations (Bsl.) | 0.007 | | | |
| | (0.039) | | | |
| Hope (Bsl.) | | 0.125*** | | |
| | | (0.030) | | |
| Agency (Bsl.) | | | 0.142*** | |
| | | | (0.030) | |
| Pathways (Bsl.) | | | | 0.029 |
| | | | | (0.024) |
| Constant | -0.920*** | -0.409** | -0.256 | -0.458*** |
| | (0.194) | (0.168) | (0.180) | (0.155) |
| Baseline controls | Yes | Yes | Yes | Yes |
| Observations | 618 | 618 | 618 | 618 |
| R-squared | 0.043 | 0.095 | 0.096 | 0.057 |
| Lincom: (Video + Sav. Cal.) + | -0.307 | -0.311* | -0.306* | -0.225 |
| (Female × (Video + Sav. Cal.)) | (0.227) | (0.170) | (0.165) | (0.181) |

*Notes:* The table reports the results of an ANCOVA specification to assess the heterogeneous treatment effect of gender (variable Female, which takes a value of 1 when the subject is female; 0 otherwise) on hope, agency and pathways. The two treatment variables are video plus placebo calendar (*Video + Plac. Cal.*) and video plus savings reminder calendar (*Video + Sav. Cal.*). The controls are Female (defined above); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Full table showing coefficients of controls is presented in Appendix C. Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

adverse external shocks. Unattainability of meeting such commitments likely frustrated participants' sense of Agency, reducing their overall Hope. Our finding is then explained by females in our sample facing greater obstacles in terms of turning their Aspirations into reality compared to male participants. This pattern is consistent with the broader gender disparities discussed above: women's limited access to resources and decision-making power may have made it harder for them to act on their aspirations, especially when external shocks compounded these constraints.

**Initial total savings.** Table 7 presents ANCOVA results of heterogenous treatment effects of total savings at baseline on Aspirations. An enhancing effect of both the Video + Plac. Cal. and the Video + Sav. Cal., treatments on Aspirations is seen for those who saved more at baseline. Specifically, we find that every 1000 Ghana cedis increase in total savings at baseline is associated with the Video + Plac. Cal. and the Video + Sav. Cal. treatment effects on aspirations being higher by 0.086 standard deviations (see Column 1 of Table 7). This result echoes related literature (see Gehrke et al., 2023) that finds that treatments effects on educational aspirations are higher for those that already received better grades at baseline; the analogy being that our effects are higher for those who already saved more at baseline. As for hope and its components, we fail to detect any significance regarding the heterogenous effects regarding initial savings.

## 5.4 Savings outcomes and heterogeneous treatment effects

**Saving for old age — extensive margin.** The ANCOVA results presented in Table 8 show the effects of the Video + Plac. Cal. and Video + Sav. Cal. treatments on the probability that subjects save for old age. The findings reveal that subjects who received the Video + Sav. Cal. treatment were 0.172 times more likely to save for old age compared to the control group (Column 1). This suggests that the combination of the video and the savings calendar had a positive impact on encouraging individuals to save for their retirement in the long run.

To further explore potential heterogeneous treatment effects, Column 2 presents ANCOVA results examining the interactions between gender and the treatments on saving for old age. The analysis finds no significant heterogeneous treatment effects of gender, i.e., we find no evidence that the effectiveness of our intervention in promoting saving for old age differed between male and female participants. On the other hand, Column 3 presents ANCOVA results investigating the heterogeneous treatment effect of total baseline savings on saving for old age. The findings indicate no statistically significant heterogeneous treatment effects. In other words, we do not find evidence that the impact of our interventions on encouraging saving for old age varied significantly based on participants' initial savings levels.

Table 7: Heterogeneous treatment effect by baseline total savings

| VARIABLES | (1) Aspirations | (2) Hope | (3) Agency | (4) Pathways |
|---|---|---|---|---|
| Video + Plac. Cal. | -0.048 | 0.168* | 0.166 | 0.071 |
| | (0.119) | (0.096) | (0.101) | (0.086) |
| Video + Sav. Cal. | -0.085 | -0.007 | -0.047 | 0.041 |
| | (0.121) | (0.100) | (0.104) | (0.093) |
| Total savings (Bsl.) | -0.023*** | 0.003 | 0.004 | -0.006 |
| | (0.008) | (0.009) | (0.010) | (0.006) |
| Total savings (Bsl.) × (Video + Plac. Cal.) | 0.072** | 0.013 | 0.030 | -0.006 |
| | (0.032) | (0.021) | (0.022) | (0.018) |
| Total savings (Bsl.) × (Video + Sav. Cal.) | 0.086** | 0.049 | 0.047 | 0.033 |
| | (0.035) | (0.036) | (0.040) | (-0.006) |
| Aspirations (Bsl.) | -0.002 | | | |
| | (0.043) | | | |
| Hope (Bsl.) | | 0.136*** | | |
| | | (0.038) | | |
| Agency (Bsl.) | | | 0.164*** | |
| | | | (0.038) | |
| Pathways (Bsl.) | | | | 0.017 |
| | | | | (0.028) |
| Constant | | -0.399* | -0.098 | -0.606*** |
| | | (0.221) | (0.238) | (0.192) |
| Baseline controls | Yes | Yes | Yes | Yes |
| Observations | 367 | 367 | 367 | 367 |
| R-squared | 0.036 | 0.092 | 0.127 | 0.040 |
| Lincom: (Video + Sav. Cal.) + | 0.001 | XXX | 0.000 | 0.074 |
| (Total savings (Bsl.) × (Video + Sav. Cal.)) | (0.111) | | (0.092) | (0.083) |

*Notes:* The table reports the results of an ANCOVA specification to assess the heterogeneous treatment effect of gender (variable Female, which takes a value of 1 when the subject is female; 0 otherwise) on hope, agency and pathways. The two treatment variables are video plus placebo calendar (*Video + Plac. Cal.*) and video plus savings reminder calendar (*Video + Sav. Cal.*). The controls are Female (defined above); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Full table showing coefficients of controls is presented in Appendix C. Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

**Pension savings — intensive margin.** Our analysis reveals a marginally significant and positive effect of the Video + Sav. Cal. treatment on pension saving intensity with PPT. Farmers who received this treatment increased their pension savings by approximately GH 39 when we add control variables (see Column 4 of Table 8). To contextualize this amount, a day's wage for a farm laborer was around GH 50 at the time of the intervention.

Table 8: Impact of Video + Plac. Cal. and Video + Sav. Cal. treatments on saving outcomes

| | Saving for old age (Endl.) | | | PPT pension savings | | |
|---|---|---|---|---|---|---|
| VARIABLES | (1) | (2) | (3) | (4) | (5) | (6) |
| Video + Plac. Cal. | 0.004 | -0.017 | 0.110 | 28.279 | 40.488 | -2.950 |
| | (0.088) | (0.094) | (0.124) | (23.411) | (27.612) | (26.519) |
| Video + Sav. Cal. | 0.172* | 0.151 | 0.084 | 39.201* | 36.069 | -17.995 |
| | (0.096) | (0.104) | (0.127) | (22.855) | (24.809) | (27.888) |
| Saving for old age (Bsl.) | 0.334*** | 0.333*** | 0.352*** | | | |
| | (0.079) | (0.079) | (0.103) | | | |
| Female | | 0.044 | | | 15.773 | |
| | | (0.189) | | | (40.474) | |
| Female × (Video + Plac. Cal.) | | 0.133 | | | -66.336 | |
| | | (0.241) | | | (48.625) | |
| Female × (Video + Sav. Cal.) | | 0.140 | | | 5.793 | |
| | | (0.278) | | | (63.243) | |
| Total savings (Bsl.) | | | -0.024* | | | 1.423 |
| | | | (0.014) | | | (3.993) |
| Total savings (Bsl.) × (Video + Plac. Cal.) | | | 0.014 | | | 7.056 |
| | | | (0.027) | | | (15.894) |
| Total savings (Bsl.) × (Video + Sav. Cal.) | | | 0.057 | | | 42.971** |
| | | | (0.035) | | | (17.872) |
| Constant | 1.642*** | 1.673*** | 1.765*** | 35.637 | 29.936 | 93.168 |
| | (0.239) | (0.244) | (0.322) | (47.285) | (47.250) | (64.151) |
| Baseline controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 618 | 618 | 367 | 235 | 235 | 153 |
| R-squared | 0.051 | 0.051 | 0.065 | 0.027 | 0.034 | 0.170 |

*Notes:* Columns (1-3) report the results of an ANCOVA specification to assess treatments effects, the heterogeneous treatment effect of gender (variable Female, which equals 1 when the subject is female; 0 otherwise), and the heterogeneous treatments effects of total savings at baseline (variable Total savings (Bsl)) on savings for old age (which equals to "1" when the subject saves for old age and "0" otherwise). Columns (4-6) instead report those effects on on PPT pension savings intensity (recorded in Ghana cedis). The two treatment variables are video plus placebo calendar (Video + Plac. Cal.) and video plus savings reminder calendar (Video + Sav. Cal.). Controls are variables that were unbalanced at the baseline. These are Age (the age of the respondent); Migration status (a dummy variable that takes a value of 1 if a person is a first-generation migrant in the community where they are living now; "0" otherwise, i.e., native or second generation onwards); Salience of old age (a variable that takes the value of 1 if a respondent often think about old age;0 otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (a variable that takes a value of 1 if the respondent has had to sell assets due to distress from debt; 0 otherwise); Natural shocks (a variable that takes the value of 1 if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; 0 otherwise).Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Due to non-responses, the number of observations used for these estimates is substantially lower. To ensure that the non-responses were random and not systematically related to the outcome variable, we employ a Heckman regression model. This approach allows us to assess whether the unobserved factors influencing the selection process (i.e., responding to the PPT savings question) are significantly related to the unobserved factors affecting the intensity of PPT savings. The results indicate no significant correlation between the error terms of the selection equation and the outcome equation (see Table E.1 in Appendix E). Furthermore, the inclusion of significant predictors of responding to the PPT savings question from the Heckman selection model does not substantially alter

the estimated treatment effects on PPT savings (see Table E.2 in Appendix E), providing evidence of the robustness of our finding. We further explore heterogeneity of the treatment effect on PPT pension savings based on gender and total savings at baseline. We find no heterogeneous treatment effects of gender on PPT pension savings intensity (see Column 5 of Table 8).

However, we do observe the treatment effect of the Video + Sav. Cal. on the average amount of pension savings to be significantly higher among smallholders who had higher initial levels of total savings. According to our estimates, these farmers saved approximately GH 43 more (see Column 6). Thus, the treatment seems to have reinforced existing savings behavior: while the treatment effect on the extensive margin of savings (i.e., fraction of farmers who save for old age, see Table 8) does not vary by baseline savings level, the intensive margin is stronger for those with higher baseline savings. This result mirrors our finding that higher initial levels of total savings predict higher marginal effects on Aspirations and is once again reminiscent of the findings of Gehrke et al. (2023) regarding educational outcomes. They observed substantial heterogeneity in treatment effects with respect to baseline student performance: specifically, their intervention increased schooling efforts among high-performing students, while treated low-performing students actually reduced their educational investments.

# 6 Discussion and conclusion

This study evaluates the effectiveness of a motivational video and a savings calendar designed to encourage smallholder cocoa farmers in rural Ghana to save more for old age. The two behavioral interventions were grounded in the literature on aspirations and hope, aiming to address both psychological and practical barriers to long-term savings. Our findings provide important insights into how such interventions can influence savings behavior, particularly in contexts where formal social safety nets are weak or absent.

The most significant result is that the combination of the video and the savings calendar led to a statistically significant increase in the fraction of farmers saving for old age in the long run. Farmers in this group were 17.2% more likely to save for old age compared to the control group. This suggests that the intervention successfully motivated farmers to take action toward securing their financial future, even if the psychological

effects (such as aspirations and hope) were not uniformly sustained.

The intervention's impact on the amount saved was particularly pronounced among farmers who already had higher levels of savings at baseline. Those with greater initial savings increased their pension contributions by approximately GH¢43 (about one day's wage for a farm laborer at the time of the study). This aligns with findings from related literature (e.g., Gehrke et al., 2023), where interventions tend to reinforce existing positive behaviors rather than uniformly lifting all participants.

While the intervention increased overall savings participation, its effects were uneven: although the savings impact did not differ between women and men, women benefited less than men in terms of agency and hope. The rigid commitment implied by writing down savings goals may have backfired for women, who probably face greater structural constraints (e.g., limited access to resources, decision-making power, and social support).

The video intervention alone had a positive long-term effect on hope and agency, suggesting that aspirational messaging can foster a sense of control over one's financial future. However, this did not always translate into increased savings unless paired with a practical tool (the savings calendar). This underscores the importance of combining motivational and structural supports to achieve behavioral change.

While the video successfully raised aspirations in the short term, its effects on aspirations faded over time, which is in line with pattern observed in other studies (e.g., Rojas Valdes et al., 2022). To sustain long-term behavioral change, interventions should incorporate ongoing engagement mechanisms, such as periodic reminders, peer support groups, or adaptive goal-setting tools. The savings calendar's success in increasing participation suggests that visual and tangible tools can help bridge the gap between intentions and actions.

The finding that farmers with higher initial savings responded more strongly to the intervention implies that one-size-fits-all approaches may not be optimal. Policymakers and practitioners should consider segmenting interventions, for example, by offering additional support (e.g., financial literacy training, access to credit) to those with lower baseline savings to help them overcome structural barriers.

The negative impact on women's sense of agency and hope is a critical concern. Future interventions should be designed with gender-specific adaptations, such as Flexible goal-setting (e.g., allowing adjustments for unforeseen shocks), targeted messaging

that addresses women's unique constraints (e.g., lack of control over household finances) and/or complementary resources (e.g., access to women's savings groups, microcredit, or land rights programs).

The study suggests that psychological interventions alone may not be sufficient to drive lasting behavioral change. While the video and savings calendar combination proved effective in increasing pension savings participation, its heterogeneous effects, especially the gender effects, suggests that successful interventions must address both internal (e.g., aspirations, agency) and external (e.g., structural constraints, gender disparities) barriers.

While this study provides valuable insights, several limitations should be acknowledged. First, the relatively small sample size and the focus on cocoa farmers in Ghana may limit the broader applicability of the findings. Future research should explore similar interventions in other contexts and with larger samples. Second, the study period coincided with the COVID-19 pandemic, which may have disproportionately affected vulnerable groups, such as women. Longitudinal studies could examine how such shocks interact with intervention effects over time. Finally, while we observe that the savings calendar increased participation, the precise mechanisms (e.g., increased salience, social accountability, or habit formation) remain unclear. Further qualitative and quantitative research could unpack these pathways.

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# Appendices

## A Experiment details

### A.1 Calendar treatments

Figure A.1 depicts the placebo as well as the treatment (savings) calendars.

(a) Placebo calendar

(b) Savings calendar

Figure A.1: "Placebo calendar" (left frame) and "Savings calendar" (right frame).

### A.2 Video treatment

Below we provide the content of our treatment video, while Figure A.2 provides a screenshot of the video.

***[Scene 1.]***

An older pension recipient (in front their house) talks about his problems back in times past and how his saving for old age using a pension product made a big difference in a quite generic way. This older pension recipient has comparable demographic and socio-economic characteristics as the smallholder farmers involved in the study.

***[Narration in Scene 1.]***

Figure A.2: Screenshot from the video (Scene 2)

**Old man:** I remember when I was a young cocoa farmer back in the day with a small farm. My earnings then were just enough to see me through the season until I sold my produce again after harvesting. I had a wife and young children and because of that I thought I couldn't afford to put some monies down for retirement. My uncle, a colleague advised me to save for my retirement, and I'm glad I did. I was concerned I wouldn't have enough money to live on, talk of save for retirement that seemed so far off, but he entreated me to put away some money no matter how small. He said small regular contributions were better than nothing at all and he highlighted on the importance of consistency. I had another uncle who had not saved at all, and because of that, he still had to work even when he was well into his old age to fend for his family. I remember how unhappy he was, and how a once vibrant man like him, resorted to begging and odd jobs in his later years. With that picture in mind, I resolved to save for my retirement, and I can confidently say that it is one of the best decisions of my life. I married quite late, my daughter is still in secondary school and my son is learning a trade. My pension can take care of my needs and that of my family. Look, I can afford to renovate my house even in my retirement. It is not too late to save for retirement. A little retirement savings can improve your life when you are old.

***[Scene 2.]***

A group meeting (cocoa cooperative) where a discussion about old age/pensions is the focus. It featured a young, middle-aged and an older cocoa farmer, where the middle-aged farmers bring up their concerns about old-age, the older farmers are addressing these concerns, referring to how saving in a pension plan made a difference for them and the younger farmers are corrected when saying that these are matters that are too far away for them and thus do not concern them at this stage. All age groups featured men and women. A widow who is a pension recipient also featured among the older age groups. Storylines

included personal information such as how becoming a widow influenced their perception of saving for old age.

***[Narration in Scene 2.]***

**Group leader**: As we announced at our last meeting, we received a letter indicating that people who know about pension will come and discuss pensions with us. Today, we will be discussing the importance of pension in general and why we should all join a pension scheme

**Young Man**: I am curious about why Mr. Addo emphasized the importance of people joining a pension scheme, because, as we are all aware, young people have enough strength to work and save for their retirement later. Why must we burden ourselves with an event that will happen in the very distant future. I think we have more pressing needs to attend to today, and honestly, pension should not be part of it.

**Old man:** I once was as young as you are. I actually wish I had started earlier than I did. At your current stage, retirement may appear to you as a far-fetched event, which really is not the case. The nature of our job permits seasonal income, thus your income is periodic and irregular, as opposed to our counterparts in other sectors. It is important to consider all these factors when planning for your retirement and save diligently. Retirement is not so far off as you may think. Luckily for me, I realized the importance of preparing for this inevitable stage in life and took active measures toward it. Thankfully, I am enjoying the benefit thereof today. I have enough funds to cater for my health needs as well other pressing financial challenges, considering that I am my family head. I did not envision all these responsibilities and challenges when I was saving for my retirement but here, I am now assuming these roles anyway. Id advice you to take this pension education seriously and save actively for it now. When you are old and frail, you will be glad you did.

**Widow**: I agree with what Mr Atakorah just said. My late husband enrolled on a personal pension scheme and encouraged me to join too. I made regular contribution from the money I got from selling the produce from my farm. My husband died 7 years ago. Aside the funds I received as his surviving beneficiary, I also receive a monthly pension from my own contributions. Most often than not, we women relegate such issues to our husbands. In case of the unfortunate demise of the man, we are left with little to no funds to cater for our needs. Pension is not for the preserve of men only. I will encourage all women to enroll on a personal pension plan too. It has been very helpful to me in my old age.

# B Appendix: attrition and balance tests

Table B.1: Attrition test on treatments, baseline outcomes and demographics

| VARIABLES | Attrition | VARIABLES | Attrition |
|---|---|---|---|
| Video + Plac. Cal. | -0.022 | Income | 0.000 |
| | (0.037) | | (0.000) |
| Video + Sav. Cal. | -0.041 | Total farm size | -0.001 |
| | (0.036) | | (0.001) |
| Aspirations (Bsl.) | -0.015 | Total savings (Bsl.) | -0.002 |
| | (0.018) | | (0.005) |
| Agency (Bsl.) | 0.010 | Saving for old age | 0.027 |
| | (0.018) | | (0.032) |
| Pathways (Bsl.) | 0.001 | Salience of old age | -0.428*** |
| | (0.018) | | (0.032) |
| Female | 0.004 | House size | 0.001 |
| | (0.042) | | (0.006) |
| Age | -0.001 | Elec. and electr. assets | -0.005 |
| | (0.001) | | (0.011) |
| Household size | -0.001 | Vehicles | 0.033 |
| | (0.006) | | (0.025) |
| Migration status | -0.006 | Farm equipment | 0.024 |
| | (0.034) | | (0.023) |
| Speaks Twi | -0.040 | Leadership status | -0.003 |
| | (0.041) | | (0.043) |
| Education | -0.004 | Social capital | 0.003 |
| | (0.006) | | (0.013) |
| Financial literacy | 0.014 | Financial difficulty | 0.039 |
| | (0.033) | | (0.037) |
| | | Natural shocks | 0.013 |
| | | | (0.031) |
| Constant | 0.539*** | | |
| | (0.112) | | |
| Observations | 387 | | |
| R-squared | 0.357 | | |

*Notes:* Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.2: Balance table for treatments

| Category | Variables | Obs. | Control (1) | *Video + Plac. Cal.* (2) | *Video + Sav. Cal.* (3) | (1) vs. (2) | (1) vs. (3) | (2) vs. (3) |
|---|---|---|---|---|---|---|---|---|
| **Main Outcomes** | | | | | | | | |
| | Aspirations (Bsl.) | 723 | 0.000 | 0.049 | 0.068 | -0.049 | -0.068 | -0.020 |
| | | | (0.065) | (0.052) | (0.066) | (0.083) | (0.093) | (0.084) |
| | Agency (Bsl.) | 723 | 0.000 | 0.145 | 0.121 | -0.145 | -0.121 | 0.024 |
| | | | (0.065) | (0.065) | (0.066) | (0.092) | (0.093) | (0.093) |
| | Pathways (Bsl.) | 723 | 0.000 | -0.050 | 0.131 | 0.050 | -0.131 | -0.181* |
| | | | (0.065) | (0.078) | (0.064) | (0.102) | (0.091) | (0.100) |
| **Demographics** | | | | | | | | |
| | Female | 723 | 0.145 | 0.250 | 0.194 | -0.105*** | -0.049 | 0.056 |
| | | | (0.023) | (0.028) | (0.025) | (0.036) | (0.034) | (0.038) |
| | Age | 721 | 57.043 | 55.651 | 54.923 | 1.391 | 2.119* | 0.728 |
| | | | (0.811) | (0.838) | (0.777) | (1.167) | (1.123) | (1.142) |
| | Household size | 723 | 5.723 | 5.721 | 5.540 | 0.003 | 0.183 | 0.181 |
| | | | (0.179) | (0.153) | (0.144) | (0.235) | (0.228) | (0.209) |
| | Migration status | 722 | 0.329 | 0.342 | 0.258 | -0.013 | 0.071* | 0.084** |
| | | | (0.031) | (0.031) | (0.028) | (0.043) | (0.041) | (0.041) |
| | Speaks Twi | 723 | 0.826 | 0.808 | 0.823 | 0.017 | 0.003 | -0.014 |
| | | | (0.025) | (0.025) | (0.024) | (0.036) | (0.035) | (0.035) |
| | Education | 705 | 9.180 | 9.110 | 9.527 | 0.070 | -0.347 | -0.417 |
| | | | (0.180) | (0.199) | (0.174) | (0.269) | (0.250) | (0.264) |
| | Financial literacy | 723 | 0.617 | 0.633 | 0.673 | -0.016 | -0.056 | -0.040 |
| | | | (0.032) | (0.031) | (0.030) | (0.045) | (0.044) | (0.043) |
| **Income and Savings** | | | | | | | | |
| | Income | 662 | 8.073 | 7.907 | 17.409 | 0.166 | -9.336 | -9.502 |
| | | | (0.549) | (0.566) | (7.936) | (0.790) | (8.323) | (8.097) |
| | Total farm size | 717 | 8.472 | 10.871 | 8.679 | -2.399 | -0.207 | 2.192 |
| | | | (1.031) | (1.314) | (1.371) | (1.672) | (1.729) | (1.902) |
| | Total savings (Bsl.) | 423 | 1.311 | 1.260 | 1.353 | 0.051 | -0.042 | -0.093 |
| | | | (0.295) | (0.198) | (0.214) | (0.359) | (0.359) | (0.295) |
| | Saving for old age | 723 | 0.600 | 0.575 | 0.536 | 0.025 | 0.064 | 0.039 |
| | | | (0.032) | (0.032) | (0.032) | (0.045) | (0.045) | (0.045) |
| | Salience of old age | 723 | 0.660 | 0.675 | 0.698 | -0.015 | -0.038 | -0.023 |
| | | | (0.031) | (0.030) | (0.029) | (0.043) | (0.043) | (0.042) |
| **Other Assets** | | | | | | | | |
| | House size | 723 | 4.064 | 4.025 | 4.246 | 0.039 | -0.182 | -0.221 |
| | | | (0.159) | (0.144) | (0.161) | (0.215) | (0.227) | (0.216) |
| | Electrical assets | 723 | 3.196 | 3.233 | 3.157 | -0.038 | 0.038 | 0.076 |
| | | | (0.106) | (0.105) | (0.094) | (0.149) | (0.141) | (0.140) |
| | Vehicles | 723 | 0.502 | 0.429 | 0.423 | 0.073 | 0.079 | 0.006 |
| | | | (0.044) | (0.042) | (0.040) | (0.060) | (0.059) | (0.057) |
| | Farm equipment | 723 | 0.864 | 0.887 | 0.786 | -0.024 | 0.078 | 0.101* |
| | | | (0.043) | (0.043) | (0.042) | (0.061) | (0.060) | (0.060) |
| **Social Standing** | | | | | | | | |
| | Leadership status | 723 | 0.149 | 0.150 | 0.177 | -0.001 | -0.028 | -0.027 |
| | | | (0.023) | (0.023) | (0.024) | (0.033) | (0.034) | (0.034) |
| | Social capital | 723 | 2.123 | 2.112 | 2.306 | 0.011 | -0.183* | -0.194* |
| | | | (0.073) | (0.079) | (0.076) | (0.108) | (0.106) | (0.109) |
| **Shocks** | | | | | | | | |
| | Financial difficulty | 723 | 0.140 | 0.175 | 0.198 | -0.035 | -0.057* | -0.023 |
| | | | (0.023) | (0.025) | (0.025) | (0.033) | (0.034) | (0.035) |
| | Natural shocks | 723 | 0.391 | 0.304 | 0.359 | 0.087** | 0.033 | -0.055 |
| | | | (0.032) | (0.030) | (0.031) | (0.044) | (0.044) | (0.043) |

# C Appendix: extra regression results.

Table C.1: Long-run impacts of treatments without controls

| VARIABLES | (1) Aspirations | (2) Hope | (3) Agency | (4) Pathways |
|---|---|---|---|---|
| Video + Plac. Cal. | 0.014 | 0.150** | 0.155** | 0.070 |
| | (0.081) | (0.073) | (0.075) | (0.068) |
| Video + Sav. Cal. | 0.086 | 0.073 | 0.063 | 0.065 |
| | (0.084) | (0.069) | (0.073) | (0.070) |
| Aspirations (Bsl.) | 0.022 | | | |
| | (0.039) | | | |
| Hope (Bsl.) | | 0.151*** | | |
| | | (0.032) | | |
| Agency (Bsl.) | | | 0.154*** | |
| | | | (0.030) | |
| Pathways (Bsl.) | | | | 0.052** |
| | | | | (0.026) |
| Constant | -0.499*** | -0.194*** | -0.037 | -0.346*** |
| | (0.057) | (0.052) | (0.057) | (0.048) |
| Baseline controls | No | No | No | No |
| Observations | 618 | 618 | 618 | 618 |
| R-squared | 0.003 | 0.044 | 0.054 | 0.009 |
| *Test of differences in coefficients of the two treatments* | | | | |
| F(1, 614) | XXX | 1.27 | 1.90 | 0.01 |
| Prob > F | | 0.261 | 0.169 | 0.942 |

*Notes:* The table reports the results of an ANCOVA specification to assess treatment effects on hope, agency and pathways. The two treatment variables are video plus placebo calendar (Video + Plac. Cal.) and video plus savings reminder calendar (Video + Sav. Cal.). Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.2: Heterogenous treatment effect of gender without controls

| VARIABLES | (1)<br>Aspirations | (2)<br>Hope | (3)<br>Agency | (4)<br>Pathways |
|---|---|---|---|---|
| Female | 0.190 | 0.275* | 0.313** | 0.132 |
| | (0.199) | (0.146) | (0.149) | (0.146) |
| Video + Plac. Cal. | 0.014 | 0.151* | 0.168** | 0.060 |
| | (0.089) | (0.081) | (0.085) | (0.076) |
| Video + Sav. Cal. | 0.165* | 0.147* | 0.127 | 0.132* |
| | (0.089) | (0.075) | (0.081) | (0.076) |
| Female × (Video + Plac. Cal.) | -0.090 | -0.135 | -0.199 | -0.025 |
| | (0.237) | (0.188) | (0.185) | (0.186) |
| Female × (Video + Sav. Cal.) | -0.437* | -0.449** | -0.422** | -0.357* |
| | (0.252) | (0.188) | (0.186) | (0.196) |
| Aspirations (Bsl.) | 0.017 | | | |
| | (0.038) | | | |
| Hope (Bsl.) | | 0.144*** | | |
| | | (0.032) | | |
| Agency (Bsl.) | | | 0.149*** | |
| | | | (0.030) | |
| Pathways (Bsl.) | | | | 0.048* |
| | | | | (0.025) |
| Constant | -0.523*** | -0.228*** | -0.077 | -0.363*** |
| | (0.059) | (0.056) | (0.062) | (0.051) |
| Baseline controls | No | No | No | No |
| Observations | 618 | 618 | 618 | 618 |
| R-squared | 0.011 | 0.054 | 0.063 | 0.017 |
| *Lincom:* | | | | |
| (Video + Sav. Cal.) + | -0.302* | -0.311* | -0.295* | -0.225 |
| Female × (Video + Sav. Cal.) | (0.172) | (0.170) | (0.168) | (0.181) |

*Notes:* The table reports ANCOVA estimates of heterogeneous treatment effects by gender without additional controls. Female equals 1 if the subject is female. Treatment variables are video plus placebo calendar and video plus savings reminder calendar. Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.3: Heterogenous treatment effect of initial total savings without controls

| VARIABLES | (1)<br>Aspirations | (2)<br>Hope | (3)<br>Agency | (4)<br>Pathways |
|---|---|---|---|---|
| Total savings (Bsl.) | -0.025*** | 0.007 | 0.006 | -0.001 |
| | (0.007) | (0.010) | (0.011) | (0.006) |
| Video + Plac. Cal. | -0.063 | 0.168* | 0.170* | 0.060 |
| | (0.116) | (0.098) | (0.103) | (0.086) |
| Video + Sav. Cal. | -0.078 | 0.014 | -0.013 | 0.036 |
| | (0.117) | (0.100) | (0.106) | (0.092) |
| Total savings (Bsl.) × (Video + Plac. Cal.) | 0.070** | 0.011 | 0.036* | -0.013 |
| | (0.030) | (0.020) | (0.021) | (0.017) |
| Total savings (Bsl.) × (Video + Sav. Cal.) | 0.072** | 0.035 | 0.036 | 0.020 |
| | (0.035) | (0.038) | (0.042) | (0.029) |
| Aspirations (Bsl.) | -0.003 | | | |
| | (0.042) | | | |
| Hope (Bsl.) | | 0.140*** | | |
| | | (0.039) | | |
| Agency (Bsl.) | | | 0.171*** | |
| | | | (0.038) | |
| Pathways (Bsl.) | | | | 0.017 |
| | | | | (0.028) |
| Constant | -0.416*** | -0.190*** | -0.041 | -0.313*** |
| | (0.078) | (0.068) | (0.077) | (0.055) |
| Baseline controls | No | No | No | No |
| Observations | 367 | 367 | 367 | 367 |
| R-squared | 0.013 | 0.054 | 0.091 | 0.005 |
| *Lincom:* | | | | |
| (Video + Sav. Cal.) + Total savings (Bsl.) | -0.006 | XXX | 0.023 | 0.056 |
| × (Video + Sav. Cal.) | (0.107) | | (0.093) | (0.082) |

*Notes:* The table reports ANCOVA estimates of heterogeneous treatment effects by initial total savings without additional controls. Treatment variables are video plus placebo calendar and video plus savings reminder calendar. Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.4: Impact of two treatments on saving outcomes without controls

| | Saving for old age (Endl.) | | | PPT pension savings | | |
|---|---|---|---|---|---|---|
| VARIABLES | (1) | (2) | (3) | (4) | (5) | (6) |
| Video + Plac. Cal. | 0.018 | -0.014 | 0.131 | 26.039 | 38.337 | 12.386 |
| | (0.087) | (0.094) | (0.121) | (22.701) | (27.906) | (25.289) |
| Video + Sav. Cal. | 0.195** | 0.162 | 0.128 | 37.405* | 34.683 | -13.040 |
| | (0.096) | (0.102) | (0.129) | (21.471) | (23.544) | (27.861) |
| Saving for old age (Bsl.) | 0.340*** | 0.345*** | 0.355*** | | | |
| | (0.077) | (0.077) | (0.098) | | | |
| Female | | 0.003 | | | 11.083 | |
| | | (0.187) | | | (34.964) | |
| Female × (Video + Plac. Cal.) | | 0.128 | | | -61.308 | |
| | | (0.243) | | | (44.825) | |
| Female × (Video + Sav. Cal.) | | 0.154 | | | 9.283 | |
| | | (0.274) | | | (60.554) | |
| Total savings (Bsl.) | | | -0.020 | | | 3.616 |
| | | | (0.013) | | | (2.413) |
| Total savings (Bsl.) × (Video + Plac. Cal.) | | | 0.015 | | | 1.249 |
| | | | (0.025) | | | (14.422) |
| Total savings (Bsl.) × (Video + Sav. Cal.) | | | 0.045 | | | 38.905** |
| | | | (0.034) | | | (17.636) |
| Constant | 1.807*** | 1.804*** | 1.827*** | 71.768*** | 70.417*** | 66.512*** |
| | (0.080) | (0.084) | (0.099) | (10.069) | (10.564) | (14.547) |
| Baseline controls | No | No | No | No | No | No |
| Observations | 618 | 618 | 367 | 235 | 235 | 153 |
| R-squared | 0.038 | 0.041 | 0.045 | 0.011 | 0.019 | 0.139 |

*Notes:* Columns (1–3) report ANCOVA estimates for saving for old age (Endline). Columns (4–6) report effects on PPT pension savings (in Ghana cedis). Treatment variables are video plus placebo calendar (Video + Plac. Cal.) and video plus savings reminder calendar (Video + Sav. Cal.). Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.5: Full table – Short- and long-term treatment effects on aspirations, hope, agency and pathways

| VARIABLES | (1) Aspirations (Mdl.) | (2) Aspirations (Endl.) | (3) Hope (Endl.) | (4) Agency (Endl.) | (5) Pathways (Endl.) |
|---|---|---|---|---|---|
| Video | 0.120* | | | | |
| | (0.070) | | | | |
| Video + Plac. Cal. | | 0.017 | 0.142** | 0.147** | 0.064 |
| | | (0.082) | (0.072) | (0.075) | (0.068) |
| Video + Sav. Cal. | | 0.071 | 0.058 | 0.042 | 0.057 |
| | | (0.084) | (0.068) | (0.073) | (0.070) |
| Aspirations (Bsl.) | 0.518*** | 0.012 | | | |
| | (0.075) | (0.039) | | | |
| Hope (Bsl.) | | | 0.132*** | | |
| | | | (0.030) | | |
| Agency (Bsl.) | | | | 0.147*** | |
| | | | | (0.030) | |
| Pathways (Bsl.) | | | | | 0.033 |
| | | | | | (0.024) |
| Female | 0.069 | 0.007 | 0.101 | 0.114 | 0.040 |
| | (0.091) | (0.089) | (0.073) | (0.070) | (0.078) |
| Age | -0.003 | -0.001 | -0.004* | -0.003 | -0.005** |
| | (0.003) | (0.003) | (0.002) | (0.003) | (0.002) |
| Migration status | -0.137* | 0.020 | -0.005 | -0.078 | 0.101 |
| | (0.071) | (0.072) | (0.064) | (0.068) | (0.062) |
| Salience of old age | 0.013 | 0.273*** | 0.229*** | 0.207*** | 0.182*** |
| | (0.087) | (0.080) | (0.067) | (0.075) | (0.064) |
| Farm equipment | -0.054 | 0.039 | 0.120*** | 0.111** | 0.090** |
| | (0.066) | (0.053) | (0.044) | (0.044) | (0.045) |
| Social capital | -0.016 | 0.076** | 0.033 | 0.030 | 0.036 |
| | (0.027) | (0.030) | (0.024) | (0.025) | (0.024) |
| Financial difficulty | -0.012 | -0.003 | 0.048 | 0.004 | 0.109 |
| | (0.084) | (0.088) | (0.077) | (0.076) | (0.081) |
| Natural shocks | -0.213*** | -0.069 | -0.028 | -0.063 | 0.018 |
| | (0.082) | (0.070) | (0.059) | (0.062) | (0.059) |
| Constant | 0.382* | -0.844*** | -0.335** | -0.181 | -0.407*** |
| | (0.195) | (0.196) | (0.164) | (0.177) | (0.149) |
| Observations | 618 | 618 | 618 | 618 | 618 |
| R-squared | 0.282 | 0.035 | 0.085 | 0.088 | 0.048 |

*Notes:* Column (1) reports the short-term (midline) video treatment effect on aspirations. Columns (2)–(5) report long-term (endline) treatment effects on aspirations, hope, agency, and pathways. Controls are variables that were unbalanced at the baseline. These are Female (equals "1" when the subject is female; "0" otherwise); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.6: Heterogeneous treatment effect of gender on psychological outcomes

| VARIABLES | (1)<br>Aspirations (Endl.) | (2)<br>Hope (Endl.) | (3)<br>Agency (Endl.) | (4)<br>Pathways (Endl.) |
|---|---|---|---|---|
| Female | 0.224 | 0.318** | 0.346** | 0.175 |
| | (0.192) | (0.148) | (0.149) | (0.149) |
| Video + Plac. Cal. | 0.019 | 0.146* | 0.166** | 0.050 |
| | (0.089) | (0.080) | (0.084) | (0.075) |
| Video + Sav. Cal. | 0.154* | 0.139* | 0.116 | 0.122 |
| | (0.090) | (0.075) | (0.081) | (0.076) |
| Female × (Video + Plac. Cal.) | -0.111 | -0.119 | -0.181 | -0.010 |
| | (0.229) | (0.188) | (0.181) | (0.190) |
| Female × (Video + Sav. Cal.) | -0.461* | -0.451** | -0.422** | -0.347* |
| | (0.242) | (0.187) | (0.184) | (0.198) |
| Aspirations (Bsl.) | 0.007 | | | |
| | (0.039) | | | |
| Hope (Bsl.) | | 0.125*** | | |
| | | (0.030) | | |
| Agency (Bsl.) | | | 0.142*** | |
| | | | (0.030) | |
| Pathways (Bsl.) | | | | 0.029 |
| | | | | (0.024) |
| Age | -0.000 | -0.003 | -0.002 | -0.005** |
| | (0.003) | (0.002) | (0.002) | (0.002) |
| Migration status | 0.020 | -0.005 | -0.078 | 0.101 |
| | (0.073) | (0.064) | (0.068) | (0.062) |
| Salience of old age | 0.275*** | 0.231*** | 0.209*** | 0.184*** |
| | (0.080) | (0.067) | (0.075) | (0.064) |
| Farm equipment | 0.047 | 0.126*** | 0.116*** | 0.096** |
| | (0.053) | (0.044) | (0.044) | (0.045) |
| Social capital | 0.077** | 0.034 | 0.031 | 0.036 |
| | (0.030) | (0.024) | (0.025) | (0.024) |
| Financial difficulty | -0.001 | 0.051 | 0.006 | 0.111 |
| | (0.089) | (0.076) | (0.076) | (0.081) |
| Natural shocks | -0.069 | -0.029 | -0.063 | 0.016 |
| | (0.070) | (0.059) | (0.062) | (0.059) |
| Constant | -0.920*** | -0.409** | -0.256 | -0.458*** |
| | (0.194) | (0.168) | (0.180) | (0.155) |
| Observations | 618 | 618 | 618 | 618 |
| R-squared | 0.043 | 0.095 | 0.096 | 0.057 |

*Notes:* Column (1) reports the short-term (midline) video treatment effect on aspirations. Columns (2)–(5) report long-term (endline) treatment effects on aspirations, hope, agency, and pathways. Controls are variables that were unbalanced at the baseline. These are Female (equals "1" when the subject is female; "0" otherwise); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.7: Heterogeneous treatment effect of initial savings on psychological outcomes

| VARIABLES | (1)<br>Aspirations (Endl.) | (2)<br>Hope (Endl.) | (3)<br>Agency (Endl.) | (4)<br>Pathways (Endl.) |
|---|---|---|---|---|
| Total savings (Bsl.) | -0.023*** | 0.003 | 0.004 | -0.006 |
| | (0.008) | (0.009) | (0.010) | (0.006) |
| Video + Plac. Cal. | -0.048 | 0.168* | 0.166 | 0.071 |
| | (0.119) | (0.096) | (0.101) | (0.086) |
| Total savings (Bsl.) × (Video + Plac. Cal.) | 0.072** | 0.013 | 0.030 | -0.006 |
| | (0.032) | (0.021) | (0.022) | (0.018) |
| Video + Sav. Cal. | -0.085 | -0.007 | -0.047 | 0.041 |
| | (0.121) | (0.100) | (0.104) | (0.093) |
| Total savings (Bsl.) × (Video + Sav. Cal.) | 0.086** | 0.049 | 0.047 | 0.033 |
| | (0.035) | (0.036) | (0.040) | (0.029) |
| Aspirations (Bsl.) | -0.002 | | | |
| | (0.043) | | | |
| Hope (Bsl.) | | 0.136*** | | |
| | | (0.038) | | |
| Agency (Bsl.) | | | 0.164*** | |
| | | | (0.038) | |
| Pathways (Bsl.) | | | | 0.017 |
| | | | | (0.028) |
| Female | -0.079 | 0.149 | 0.182** | 0.025 |
| | (0.128) | (0.093) | (0.086) | (0.097) |
| Age | 0.007* | -0.002 | -0.004 | 0.001 |
| | (0.004) | (0.003) | (0.003) | (0.003) |
| Migration status | -0.025 | 0.040 | -0.008 | 0.079 |
| | (0.099) | (0.085) | (0.088) | (0.080) |
| Salience of old age | 0.231** | 0.233*** | 0.212** | 0.152** |
| | (0.114) | (0.088) | (0.099) | (0.076) |
| Farm equipment | 0.009 | 0.085 | 0.074 | 0.078 |
| | (0.071) | (0.055) | (0.054) | (0.055) |
| Social capital | 0.030 | 0.005 | 0.023 | -0.013 |
| | (0.040) | (0.034) | (0.035) | (0.030) |
| Financial difficulty | 0.082 | 0.126 | 0.079 | 0.170* |
| | (0.113) | (0.092) | (0.091) | (0.092) |
| Natural/biological disaster | -0.041 | -0.021 | -0.094 | 0.091 |
| | (0.092) | (0.073) | (0.076) | (0.071) |
| Constant | -1.062*** | -0.399* | -0.098 | -0.606*** |
| | (0.281) | (0.221) | (0.238) | (0.192) |
| Observations | 367 | 367 | 367 | 367 |
| R-squared | 0.036 | 0.092 | 0.127 | 0.040 |

*Notes:* Column (1) reports the short-term (midline) video treatment effect on aspirations. Columns (2)–(5) report long-term (endline) treatment effects on aspirations, hope, agency, and pathways. Controls are variables that were unbalanced at the baseline. These are Female (equals "1" when the subject is female; "0" otherwise); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.8: Full table - Treatment effects on savings results

| | Saving for old age (Endl.) | | | PPT pension savings intensity | | |
|---|---|---|---|---|---|---|
| VARIABLES | (1) | (2) | (3) | (4) | (5) | (6) |
| Video + Plac. Cal. | 0.004 | -0.017 | 0.110 | 28.279 | 40.488 | -2.950 |
| | (0.088) | (0.094) | (0.124) | (23.411) | (27.612) | (26.519) |
| Video + Sav. Cal. | 0.172* | 0.151 | 0.084 | 39.201* | 36.069 | -17.995 |
| | (0.096) | (0.104) | (0.127) | (22.855) | (24.809) | (27.888) |
| Female | 0.150 | 0.044 | 0.300** | -7.963 | 15.773 | 12.059 |
| | (0.107) | (0.189) | (0.148) | (23.584) | (40.474) | (32.175) |
| Female × (Video + Plac. Cal.) | | 0.133 | | | -66.336 | |
| | | (0.241) | | | (48.625) | |
| Female × (Video + Sav. Cal.) | | 0.140 | | | 5.793 | |
| | | (0.278) | | | (63.243) | |
| Total savings (Bsl.) | | | -0.024* | | | 1.423 |
| | | | (0.014) | | | (3.993) |
| Total savings (Bsl.) × (Video + Plac. Cal.) | | | 0.014 | | | 7.056 |
| | | | (0.027) | | | (15.894) |
| Total savings (Bsl.) × (Video + Sav. Cal.) | | | 0.057 | | | 42.971** |
| | | | (0.035) | | | (17.872) |
| Saving for old age (Bsl.) | 0.334*** | 0.333*** | 0.352*** | | | |
| | (0.079) | (0.079) | (0.103) | | | |
| Age | 0.0004 | 0.0001 | -0.002 | 0.079 | 0.113 | -0.015 |
| | (0.003) | (0.003) | (0.004) | (0.653) | (0.645) | (1.020) |
| Migration status | -0.107 | -0.107 | 0.300** | 34.861 | 34.356 | 12.059 |
| | (0.083) | (0.083) | (0.148) | (26.881) | (26.713) | (32.175) |
| Salience of old age (Bsl.) | 0.051 | 0.051 | -0.047 | 10.078 | 9.958 | 33.092 |
| | (0.103) | (0.103) | (0.118) | (21.461) | (21.439) | (43.019) |
| Farm equipment | 0.101* | 0.100* | 0.044 | -3.277 | -4.041 | 11.034 |
| | (0.058) | (0.059) | (0.134) | (16.275) | (16.499) | (24.226) |
| Social capital | 0.017 | 0.016 | 0.097 | 4.689 | 5.382 | -14.221 |
| | (0.034) | (0.035) | (0.076) | (7.128) | (7.241) | (16.002) |
| Financial difficulty | 0.118 | 0.118 | 0.016 | 6.198 | 5.512 | -14.270 |
| | (0.113) | (0.113) | (0.045) | (30.792) | (30.801) | (9.129) |
| Natural shocks | -0.046 | -0.048 | 0.089 | 16.010 | 17.285 | 13.471 |
| | (0.079) | (0.079) | (0.145) | (20.493) | (20.576) | (41.763) |
| Constant | 1.642*** | 1.673*** | 1.765*** | 35.637 | 29.936 | 93.168 |
| | (0.239) | (0.244) | (0.322) | (47.285) | (47.250) | (64.151) |
| Observations | 618 | 618 | 367 | 235 | 235 | 153 |
| R-squared | 0.051 | 0.051 | 0.065 | 0.027 | 0.034 | 0.170 |

*Notes:* Column 1 shows results of an ANCOVA specification to assess treatments effects on savings for old age (which takes a value of 1 when the subject saves for old age and 0 otherwise). Column 2-3 assess the heterogeneous treatment effect of gender (variable Female, which equals to 1 when the subject is female; 0 otherwise) and of total savings at baseline (variable Total savings (Bsl.)). Columns (4)–(6) provide counterpart results of PPT pension savings intensity (recorded in Ghana cedis). Interaction terms capture heterogeneous treatment effects by gender and baseline total savings. The controls are Female (equals "1" when the subject is female; "0" otherwise); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Huber–White robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# D Appendix: Marginal effect of savings reminder

Table D.1: Marginal effect of savings reminder calendar on hope, agency and pathways

| | Hope (Endl.) | | Agency (Endl.) | | Pathways (Endl.) | |
|---|---|---|---|---|---|---|
| VARIABLES | (1) | (2) | (3) | (4) | (5) | (6) |
| AnyVid. | 0.150** | 0.142** | 0.155** | 0.147** | 0.070 | 0.064 |
| | (0.073) | (0.072) | (0.075) | (0.075) | (0.068) | (0.068) |
| Video + Sav. Cal. | -0.077 | -0.084 | -0.093 | -0.105 | -0.005 | -0.007 |
| | (0.068) | (0.067) | (0.067) | (0.067) | (0.071) | (0.070) |
| Hope (Bsl.) | 0.151*** | 0.132*** | | | | |
| | (0.032) | (0.030) | | | | |
| Agency (Bsl.) | | | 0.154*** | 0.147*** | | |
| | | | (0.030) | (0.030) | | |
| Pathways (Bsl.) | | | | | 0.052** | 0.033 |
| | | | | | (0.026) | (0.024) |
| Female | | 0.101 | | 0.114 | | 0.040 |
| | | (0.073) | | (0.070) | | (0.078) |
| Age | | -0.004* | | -0.003 | | -0.005** |
| | | (0.002) | | (0.003) | | (0.002) |
| Migration status | | -0.005 | | -0.078 | | 0.101 |
| | | (0.064) | | (0.068) | | (0.062) |
| Salience of old age | | 0.229*** | | 0.207*** | | 0.182*** |
| | | (0.067) | | (0.075) | | (0.064) |
| Farm equipment | | 0.120*** | | 0.111** | | 0.090** |
| | | (0.044) | | (0.044) | | (0.045) |
| Social capital | | 0.033 | | 0.030 | | 0.036 |
| | | (0.024) | | (0.025) | | (0.024) |
| Financial difficulty | | 0.048 | | 0.004 | | 0.109 |
| | | (0.077) | | (0.076) | | (0.081) |
| Natural shocks | | -0.028 | | -0.063 | | 0.018 |
| | | (0.059) | | (0.062) | | (0.059) |
| Constant | -0.194*** | -0.335** | -0.037 | -0.181 | -0.346*** | -0.407*** |
| | (0.052) | (0.164) | (0.057) | (0.177) | (0.048) | (0.149) |
| Observations | 618 | 618 | 618 | 618 | 618 | 618 |
| R-squared | 0.044 | 0.085 | 0.054 | 0.088 | 0.009 | 0.048 |

*Notes:* The table reports the results of an ANCOVA specification to assess the marginal effects of having watched the video as part of either treatment group and of the savings calendar over and above that effect on hope, agency and pathways. The two treatment variables are any video treatment (AnyVid.) and video plus savings reminder calendar (Video + Sav. Cal.). Columns 1, 3 and 5 present estimates for hope, agency and pathways respectively without controls. Columns 2, 4 and 6 present estimates for hope, agency and pathways respectively with control variables that were unbalanced at the baseline. The controls are Female (equals "1" when the subject is female; "0" otherwise); Age (the age of the respondent); Migration status (equals 1 if a person is a first-generation migrant in the community where they are living now and "0" otherwise, i.e., native or second generation onwards); Salience of old age (equals "1" if a respondent often thinks about old age and "0" otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (equals "1" if the respondent had sold assets due to distress from debt; "0" otherwise); Natural shocks (equals "1" if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; "0" otherwise). Huber–White robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.2: Marginal effect of savings reminder calendar on PPT pension savings intensity

| VARIABLES | (1)<br>PPT pension savings intensity |
|---|---|
| AnyVid. | 28.279 |
| | (23.411) |
| Video + Sav. Cal. | 10.922 |
| | (29.601) |
| Female | -7.963 |
| | (23.584) |
| Age | 0.079 |
| | (0.653) |
| Migration status | 34.861 |
| | (26.881) |
| Salience of old age | 10.078 |
| | (21.461) |
| Farm equipment | -3.277 |
| | (16.275) |
| Social capital | 4.689 |
| | (7.128) |
| Financial difficulty | 6.198 |
| | (30.792) |
| Natural shocks | 16.010 |
| | (20.493) |
| Constant | 35.637 |
| | (47.285) |
| Observations | 235 |
| R-squared | 0.027 |

*Notes:* The table reports the results of a post-treatment regression specification to assess the marginal effects of having watched the video as part of either treatment group and of the savings calendar over and above that effect on PPT pension savings intensity (recorded in Ghana cedis), without controlling for outcome variables at baseline. The two treatment variables are any video treatment (AnyVid.) and video plus savings reminder calendar (Video + Sav. Cal.). Controls are variables that were unbalanced at the baseline. These include Female (which takes a value of 1 when the subject is female; 0 otherwise); Age (the age of the respondent); Migration status (a dummy variable that takes a value of 1 if a person is a first-generation migrant in the community where they are living now; "0" otherwise, i.e., native or second generation onwards); Salience of old age (a variable that takes the value of 1 if a respondent often think about old age;0 otherwise); Farm equipment (the number of different categories of farming equipment owned by respondents, focusing mainly on knapsack sprayers and mist blowers); Social capital (the number of categories of important people within the community, such as the chief, spiritual leader, farmer group leader, and farm certification manager, that respondent is connected to); Financial difficulty (a variable that takes a value of 1 if the respondent has had to sell assets due to distress from debt; 0 otherwise); Natural shocks (a variable that takes the value of 1 if a respondent experienced any flooding or pest and disease on their farms in the year before our intervention; 0 otherwise). Huber–White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# E Appendix: Robustness for PPT pension savings intensity results

Due to non-responses, the number of observations for the PPT pension savings intensity reduced substantially. To ensure that the non-responses were random and not systematically related to the outcome variable, we employed a Heckman regression model. This approach allows us to assess whether the unobserved factors influencing the selection process (responding to the PPT savings question) are significantly related to the unobserved factors affecting the intensity of PPT savings. In the selection model, we control for all variables in our descriptives table (see Table 3), except for income, total farm size, and education, which prevented convergence. The Heckman regression results show no significant correlation between the error terms of the selection equation and the outcome equation (see column 3 of Table E 1) but our treatments did not turn out to be significant. Nonetheless, we went ahead to do a post-treatment regression where we find the Video + Sav. Cal. to be marginally significant (Table 8). We further added significant predictors from Heckman stage one selection model (which were not already in our controls, i.e., house size and saving for old age) in our post-regression analysis as a robustness check. The estimated significant treatment effect identified for was not substantially altered by the inclusion of these predictors (see column 1 of Table E 2). We also find similar effects for the heterogeneous treatment effects when we add these predictors to the controls (see column 2 and 3 of Table E 2).

Table E.1: Treatment effect on pension savings using a Heckman regression model

| VARIABLES | (1)<br>PPT pension savings intensity | (2)<br>PPT savings dummy | (3) |
|---|---|---|---|
| Video + Plac. Cal. | 13.751 | | |
| | (17.288) | | |
| Video + Sav. Cal. | 16.563 | | |
| | (16.223) | | |
| Female | 6.714 | -0.090 | |
| | (18.551) | (0.149) | |
| Age | -0.245 | -0.010** | |
| | (0.519) | (0.005) | |
| Household size | | 0.015 | |
| | | (0.023) | |
| Migration status | 18.198 | 0.041 | |
| | (19.255) | (0.123) | |
| Speaks Twi | | -0.080 | |
| | | (0.149) | |
| Education | | -0.014 | |
| | | (0.022) | |
| Financial literacy | | 0.157 | |
| | | (0.123) | |
| Income | | 0.001 | |
| | | (0.000) | |
| Total farm size | | 0.005 | |
| | | (0.003) | |
| Saving for old age (Bsl.) | | 0.342*** | |
| | | (0.123) | |
| Salience of old age (Bsl.) | 7.624 | -0.189 | |
| | (16.123) | (0.135) | |
| House size | | -0.078*** | |
| | | (0.025) | |
| Elec. and electr. assets | | -0.039 | |
| | | (0.041) | |
| Vehicles | | -0.037 | |
| | | (0.097) | |
| Farm equipment | -0.502 | -0.083 | |
| | (12.336) | (0.092) | |
| Leadership status | | -0.126 | |
| | | (0.158) | |
| Social capital | 4.822 | 0.103** | |
| | (5.122) | (0.049) | |
| Financial difficulty | 4.403 | -0.087 | |
| | (22.034) | (0.150) | |
| Natural shocks | 11.825 | -0.071 | |
| | (16.103) | (0.120) | |
| Constant | 44.075 | 0.984** | |
| | (36.054) | (0.420) | |
| Athrho | | | -0.041 |
| | | | (0.181) |
| Lnsigma | | | 4.899*** |
| | | | (0.118) |
| Observations | 556 | 556 | 556 |

*Notes:* Column 1 presents the outcome model estimates on the effect of pension savings intensity which is the second stage with control variables that were unbalanced at the baseline. Column 2 presents selection model, thus, the first stage, which includes all control variables in Table 3. Column 3 presents the correlation coefficient between the selection model and the outcome model and the variability in the savings. Athrho suggests minimal correlation between the selection process and the outcome, Lnsigma indicates significant and substantial variability in the PPT savings of the respondents. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.2: Post-treatment regression estimates including significant predictors from the Heckman stage one selection model

| VARIABLES | (1)<br>PPT pension savings intensity | (2)<br>PPT pension savings intensity | (3)<br>PPT pension savings intensity |
|---|---|---|---|
| Video + Plac. Cal. | 28.193 | 39.920 | -0.097 |
| | (23.874) | (28.310) | (26.269) |
| Video + Sav. Cal. | 38.974* | 34.954 | -12.591 |
| | (22.743) | (24.614) | (28.564) |
| Female | -5.419 | 14.690 | 13.095 |
| | (24.540) | (43.394) | (31.598) |
| Female × Video + Plac. Cal. | | -62.495 | |
| | | (51.887) | |
| Female × Video + Sav. Cal. | | 10.652 | |
| | | (64.580) | |
| Total savings (Bsl.) | | | -0.442 |
| | | | (4.584) |
| Total savings (Bsl.) × Video + Plac. Cal. | | | 5.464 |
| | | | (15.105) |
| Total savings (Bsl.) × Video + Sav. Cal. | | | 42.247** |
| | | | (18.332) |
| Age | -0.014 | -0.005 | -0.420 |
| | (0.665) | (0.658) | (0.969) |
| Migration status | 34.988 | 34.721 | 38.317 |
| | (27.496) | (27.366) | (42.102) |
| Salience of old age | 8.823 | 9.027 | 14.122 |
| | (23.250) | (23.277) | (25.249) |
| Farm equipment | -4.553 | -5.472 | -17.449 |
| | (16.668) | (16.938) | (15.927) |
| Social capital | 4.351 | 4.985 | -15.134 |
| | (7.368) | (7.469) | (9.267) |
| Financial difficulty | 7.415 | 6.869 | 20.541 |
| | (30.855) | (30.860) | (42.583) |
| Natural shocks | 16.514 | 17.768 | 21.743 |
| | (20.339) | (20.427) | (27.108) |
| Saving for old age | 21.199 | 19.582 | 31.743 |
| | (22.185) | (22.346) | (20.031) |
| House size | 3.445 | 3.798 | 11.288** |
| | (4.258) | (4.308) | (4.876) |
| Constant | 15.470 | 11.264 | 48.980 |
| | (49.221) | (47.971) | (72.494) |
| Observations | 235 | 235 | 153 |
| R-squared | 0.034 | 0.041 | 0.201 |

*Notes:* Column 1 presents main treatment effects on PPT pension savings intensity. Columns 2 and 3 present heterogeneous treatment effects of gender and baseline total savings on PPT pension savings intensity, respectively. Huber–White robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# F Appendix: Robustness check psychological outcomes results

To check the robustness of our results for psychological outcomes, we used a Lasso double selection regression model (Belloni et al.2014; 2016) as different estimation technique where we include many control variables. In socio-economic and psychological studies, the decision of which covariates to include in analysis is a universal problem and has been mostly based on variables commonly cited in literature to impact particular outcome variables. Other studies routinely control for some demographic characteristics as a standard set of covariates. However, these approaches could fail in controlling for valid covariates and hence, lead to omitted variable bias problems. The lasso technique is an objective method that selects covariates / controls that have sufficient non-negligible empirical effect or impact on an outcome variable. We included 40 variables (see Table F.1) which constituted personal characteristics, socio-economic characteristics, financial status, psychological features and farming activities of respondents as controls. We also include our treatment variables as our main predictor. The lasso double selection procedure prevents over-selection of potentially spurious covariates, reduces error and increases statistical power.

We estimated a regression model of the following general form:

$$Y_{i,t=2} = \alpha X_{i,t=0} + \beta T_i + \epsilon_i. \tag{F1}$$

Here, $Y_{i,t=2}$ is the outcome variable during endline, $X_{i,t=0}$ represents a vector of independent baseline variables as controls (also presented in Table F.1) that can influence the outcome variable and $T_i$ is the treatment variable and $\epsilon_i$ is an error term.

The lasso double-selection procedure imposes restriction over the estimated coefficients of all right-hand-side variables for the original least square regression. In our case, apart from each of the outcome variables $Y_{i,t=2}$ (i.e., aspiration, hope, agency and pathway), we denote the vector of selected explanatory variables that we are interested in as $S$ while the large amount of potential explanatory variables is denoted as $X$. The double-selection regression specifically uses the following procedures:

- It runs a Lasso regression of $Y$ on $X$ and denote significant $X$ variables as $X_Y^*$
- Then, run a Lasso regression of S on X, and denote significant X variables as $X_S^*$
- Finally, it regresses Y on S, $X_Y^*$, and $X_S^*$

The intuition behind this procedure is that it detects those $X$ variables with non-negligible influence on the distribution of the impact of $S$ on $Y$, and discards those with negligible impacts.

Our results remain robust. The estimated significant treatment effect identified for was not substantially altered (see Table F.2). The results also remain robust when we repeated ANCOVA specification with the same smaller sample accounted for by the Lasso specification (see Table F.3).

Table F.1: Descriptive statistics of all controls used in Lasso specification

| No. | Variable | Obs | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|---|
| 1 | Life insurance | 723 | 0.015 | 0.122 | 0 | 1 |
| 2 | Health insurance | 723 | 0.935 | 0.247 | 0 | 1 |
| 3 | Give money to social capital | 723 | 1.124 | 1.033 | 0 | 5 |
| 4 | Receive money from social capital | 723 | 0.414 | 0.709 | 0 | 5 |
| 5 | Amount received from social capital | 723 | 35.061 | 143.587 | 0 | 2500 |
| 6 | Amount given to social capital | 723 | 101.683 | 315.815 | 0 | 5150 |
| 7 | Money invested in cocoa farm | 708 | 2360.484 | 2404.993 | 10 | 13980 |
| 8 | Time invested in cocoa farm | 711 | 156.127 | 136.729 | 2 | 1028 |
| 9 | Educational level invested in children | 723 | 1.573 | 1.61 | 0 | 7 |
| 10 | Number of children | 723 | 2.737 | 1.988 | 0 | 9 |
| 11 | Own non-bank savings account | 723 | 0.657 | 0.475 | 0 | 1 |
| 12 | Own susu savings | 723 | 0.444 | 0.497 | 0 | 1 |
| 13 | Frequency of susu savings | 723 | 0.385 | 0.487 | 0 | 1 |
| 14 | Plan for old age | 723 | 0.835 | 0.371 | 0 | 1 |
| 15 | High proportion of income on food expenditure | 724 | 0.555 | 0.497 | 0 | 1 |
| 16 | Financial satisfaction (current plan) | 723 | 9.36 | 1.213 | 3 | 10 |
| 17 | Financial satisfaction if nothing changes | 723 | 5.761 | 2.929 | 1 | 10 |
| 18 | Financial satisfaction with full adjustment | 723 | 8.982 | 1.511 | 1 | 10 |
| 19 | Use mobile phone services | 723 | 0.815 | 0.389 | 0 | 1 |
| 20 | Own mobile account | 723 | 0.668 | 0.471 | 0 | 1 |
| 21 | Own bank account | 723 | 0.578 | 0.494 | 0 | 1 |
| 22 | Migration status | 722 | 0.309 | 0.462 | 0 | 1 |
| 23 | Speaks Twi | 723 | 0.819 | 0.385 | 0 | 1 |
| 24 | Education | 705 | 9.275 | 2.833 | 0 | 20 |
| 25 | Financial literacy | 723 | 0.642 | 0.48 | 0 | 1 |
| 26 | Household size | 723 | 5.66 | 2.46 | 1 | 23 |
| 27 | Leadership status | 723 | 0.159 | 0.366 | 0 | 1 |
| 28 | Social capital | 723 | 2.183 | 1.184 | 0 | 4 |
| 29 | Financial difficulty | 723 | 0.172 | 0.377 | 0 | 1 |
| 30 | Natural shocks | 723 | 0.351 | 0.478 | 0 | 1 |
| 31 | Female | 723 | 0.196 | 0.398 | 0 | 1 |
| 32 | Age | 721 | 55.854 | 12.547 | 24 | 97 |
| 33 | Income | 662 | 11.261 | 71.293 | 0 | 1806.9 |
| 34 | Total farm size | 717 | 9.333 | 19.376 | 0 | 252 |
| 35 | Saving for old age (Bsl.) | 723 | 0.57 | 0.495 | 0 | 1 |
| 36 | Salience of old age (Bsl.) | 723 | 0.678 | 0.468 | 0 | 1 |
| 37 | House size | 723 | 4.113 | 2.407 | 1 | 15 |
| 38 | Elec. and electr. assets | 723 | 3.195 | 1.572 | 0 | 8 |
| 39 | Vehicles | 723 | 0.451 | 0.646 | 0 | 3 |
| 40 | Farm equipment | 723 | 0.845 | 0.662 | 0 | 2 |

Table F.2: LASSO results for short-term effect on Aspirations and long-term treatment effect on Aspirations and Hope including its components, Agency and Pathways

| VARIABLES | (1) Aspirations (Mdl.) | (2) Aspirations (Endl.) | (3) Hope (Endl.) | (4) Agency (Endl.) | (5) Pathways (Endl.) |
|---|---|---|---|---|---|
| Video | 0.157** | | | | |
| | (0.078) | | | | |
| Video + Plac. Cal. | | 0.039 | 0.161** | 0.166** | 0.078 |
| | | (0.086) | (0.074) | (0.080) | (0.069) |
| Video + Sav. Cal. | | 0.103 | 0.117* | 0.096 | 0.078 |
| | | (0.087) | (0.068) | (0.076) | (0.069) |
| *Selected variables* | | | | | |
| Aspirations (Bsl.) | 0.475 | | | | |
| Hope (Bsl.) | | | 0.095 | | |
| Agency (Bsl.) | | | | 0.138 | |
| Maximum life satisfaction | | | | | |
| from expected lifestyle change | | | 0.069 | | |
| Elec. and electr. assets | | | 0.062 | 0.075 | |
| Total farm size | | | 0.005 | 0.004 | |
| Salience of old age | | | 0.259 | | |
| Number of selected variables | 1 | 0 | 5 | 3 | 0 |
| Number of baseline controls | 41 | 41 | 41 | 41 | 41 |
| Constant | 0.145 | -0.431 | -1.167 | -0.233 | -0.273 |
| Observations | 634 | 548 | 548 | 548 | 548 |

*Notes:* Lasso selects variables with non-negligible effect for each outcome. For short term aspirations - Aspirations (Mdl.) - it selects only Aspirations (Bsl.). For long term aspirations - Aspirations (Endl.) - and pathways - Pathways (Endl.) -, it selects none of the controls. For long term hope - Hope (Endl.) -, it selects Hope (Bsl.), Maximum life satisfaction from expected lifestyle change, owning an electrical and electronic asset (Elec and electr assets), Total farm size and Salience of old age. For long term agency - Agency (Endl.) -, it selects, Agency (Bsl.), owning an electrical and electronic asset (Elec and electr assets) and Total farm size. Huber-White robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F.3: ANCOVA results for short-term effect on Aspirations and long-term treatment effect on Aspirations and Hope including its components, Agency and Pathways

| VARIABLES | (1) Aspirations (Mdl.) | (2) Aspirations (Mdl.) | (3) Aspirations (Endl.) | (4) Aspirations (Endl.) | (5) Hope (Endl.) | (6) Hope (Endl.) | (7) Agency (Endl.) | (8) Agency (Endl.) | (9) Pathways (Endl.) | (10) Pathways (Endl.) |
|---|---|---|---|---|---|---|---|---|---|---|
| Video | 0.157** | 0.133* | | | | | | | | |
| | (0.078) | (0.077) | | | | | | | | |
| Video + Plac. Cal. | | | 0.039 | 0.041 | 0.157** | 0.147* | 0.166** | 0.152* | 0.076 | 0.075 |
| | | | (0.086) | (0.087) | (0.077) | (0.076) | (0.082) | (0.081) | (0.069) | (0.069) |
| Video + Sav. Cal. | | | 0.112 | 0.109 | 0.092 | 0.083 | 0.088 | 0.069 | 0.070 | 0.075 |
| | | | (0.087) | (0.087) | (0.071) | (0.070) | (0.078) | (0.077) | (0.069) | (0.069) |
| Female | | 0.053 | | -0.029 | 0.103 | | 0.145* | | -0.001 | |
| | | (0.095) | | (0.096) | (0.079) | | (0.076) | | (0.082) | |
| Age | | -0.003 | | 0.004 | -0.002 | | -0.002 | | -0.002 | |
| | | (0.003) | | (0.003) | (0.002) | | (0.003) | | (0.002) | |
| Migration status | | -0.098 | | 0.023 | 0.018 | | -0.062 | | 0.126** | |
| | | (0.069) | | (0.076) | (0.068) | | (0.074) | | (0.062) | |
| Salience of old age | | 0.102 | | 0.233*** | 0.257*** | | 0.244*** | | 0.187*** | |
| | | (0.078) | | (0.082) | (0.071) | | (0.080) | | (0.064) | |
| Farm equipment | | -0.065 | | 0.014 | 0.113** | | 0.121*** | | 0.061 | |
| | | (0.063) | | (0.055) | (0.045) | | (0.046) | | (0.046) | |
| Social capital | | 0.009 | | 0.082*** | 0.017 | | 0.013 | | 0.028 | |
| | | (0.026) | | (0.032) | (0.025) | | (0.026) | | (0.025) | |
| Financial difficulty | | -0.036 | | -0.064 | 0.038 | | 0.022 | | 0.059 | |
| | | (0.084) | | (0.091) | (0.080) | | (0.080) | | (0.083) | |
| Natural shocks | | -0.159* | | -0.060 | -0.037 | | -0.067 | | 0.007 | |
| | | (0.084) | | (0.075) | (0.062) | | (0.067) | | (0.061) | |
| Hope (Bsl.) | | | | | 0.141*** | 0.128*** | | | | |
| | | | | | (0.035) | (0.034) | | | | |
| Agency (Bsl.) | | | | | | | 0.154*** | 0.146*** | | |
| | | | | | | | (0.032) | (0.032) | | |
| Pathways (Bsl.) | | | | | | | | | 0.036 | 0.024 |
| | | | | | | | | | (0.027) | (0.026) |
| Constant | 0.039 | 0.269 | -0.483*** | -1.040*** | 0.179*** | -0.426** | -0.033 | -0.238 | -0.326*** | -0.522*** |
| | (0.068) | (0.261) | (0.061) | (0.210) | (0.053) | (0.177) | (0.061) | (0.194) | (0.046) | (0.154) |
| Observations | 634 | 634 | 548 | 548 | 548 | 548 | 548 | 548 | 548 | 548 |
| R-squared | 0.202 | 0.217 | 0.003 | 0.033 | 0.041 | 0.078 | 0.055 | 0.092 | 0.006 | 0.034 |

*Notes:* Lasso selects variables with non-negligible effect for each outcome. For short term aspirations - Aspirations (Mdl.) - it selects only Aspirations (Bsl.). For long term aspirations - Aspirations (Endl.) - and pathways - Pathways (Endl.) -, it selects none of the controls. For long term hope - Hope (Endl.) -, it selects Hope (Bsl.), Maximum life satisfaction from expected lifestyle change, owning an electrical and electronic asset (Elec and electr assets), Total farm size and Salience of old age. For long term agency - Agency (Endl.) -, it selects, Agency (Bsl.), owning an electrical and electronic asset (Elec and electr assets) and Total farm size. Huber-White robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.